Robust Bayesian Inference under Partially Specified Priors*

Martin Fankhauser†

November 2026

Abstract

Set identification leaves multiple structural parameter values observationally equivalent, often generating wide identified sets and correspondingly ambiguous economic conclusions. Bayesian analysis offers a way to reduce this ambiguity by using prior information to weight these values differently, but economic reasoning may only partially identify such prior beliefs. This paper develops a robust Bayesian framework that imposes only economically meaningful restrictions on selected prior marginals while remaining agnostic about otherwise unspecified prior features. In set-identified models, the resulting posterior bounds reduce to linear optimization problems and can be approximated by reweighting simulated structural draws. Applied to oil-market SVARs, we find that demand shocks robustly dominate supply shocks in line with the literature, while the stronger conclusion that supply shocks are nearly irrelevant requires much tighter prior discipline on short-run elasticities.

Keywords: Prior Sensitivity, Robust Bayes, Imprecise Probabilities, Marginally-specified priors, Oil Demand and Supply
JEL Codes: C11, C14, C18, C32, C52, E52, E58

*We thank Massimiliano Marcellino, Francesca Molinari, Raffaella Giacomini, Toru Kitagawa, Pamela Giustinelli, Geert Mesters, Giorgio Primiceri, Nicola Pavoni, Mikkel Plagborg-Møller, Lutz Kilian, Daniel Lewis, Efrem Castelnuovo, Giovanni Pellegrino, Michele Piffer, Amilcar Velez, Jiaming Huang and Tommaso Tornese for helpful discussions and comments. All mistakes are my own.

†Bocconi University

1 Introduction

In many empirical problems, researchers are only able to set identify a parameter of interest, so that ambiguity about its value persists even asymptotically. This may arise from incomplete data, such as missing covariates, or from incomplete models, including games with multiple equilibria and sign-restricted structural models. For a recent review, see Molinari (2020). The usual object of interest is the sharp identified set, that is, the smallest set of parameter values consistent with the data and the maintained model. In many applications, however, this set remains wide, leaving substantial ambiguity about the economic question the researcher ultimately seeks to answer. A natural question is therefore whether additional information can be used to sharpen conclusions without abandoning the partial-identification principle of imposing only assumptions that can be credibly defended; see, e.g., Manski (2003).

Bayesian analysis provides a natural way to incorporate such additional information. Parameter values that are observationally equivalent receive the same likelihood, but need not receive the same prior probability. Prior information can therefore differentially weight values within the identified set and sharpen posterior conclusions beyond what is implied by the likelihood alone; see Moon and Schorfheide (2012), Gustafson (2014), Baumeister and Hamilton (2015), and Giacomini and Kitagawa (2021). The difficulty is that conventional Bayesian inference requires a fully specified prior distribution, even when the economic reasoning motivating that prior justifies only a few of its features. For example, a researcher may have good reasons to believe that smaller values of a parameter are more plausible than larger ones, without having a defensible view about the precise shape, concentration, or tail behavior of its prior distribution. In this sense, economic reasoning may only partially identify prior beliefs, just as data may only partially identify structural parameters.

This tension is especially acute in models with a high-dimensional structural parameter vector θ, where a conventional Bayesian analysis requires a full joint prior over all of its elements, while credible prior information typically concerns only a few economically

interpretable functionals $S(\theta)$, such as elasticities. This paper develops methods to bound posterior conclusions about a target of interest $T(\theta)$ over classes of priors that restrict only what the researcher can defend about $S(\theta)$, and makes three contributions.

First, we propose a framework in which the researcher imposes exactly the prior information that can be defended, and nothing more. Single-prior approaches, such as Baumeister and Hamilton (2015) or Arias and Waggoner (2024), fix every feature of the structural prior, including many that no economic argument speaks to. The full-ambiguity robust Bayesian approach of Giacomini and Kitagawa (2021) goes to the opposite extreme: it fixes nothing about the conditional structural prior beyond its support on the identified set, and thereby also discards prior knowledge the researcher may credibly hold. Our framework lies between these two poles, both of which arise as special cases: the researcher restricts selected features of the prior distribution of economically meaningful quantities $S(\theta)$, for example through probability bounds, moment conditions, or shape restrictions, and leaves all remaining features unrestricted. A useful conceptual device is to view any joint prior as a marginal distribution for $S(\theta)$ together with a completion of that marginal into a joint structural prior. Economic reasoning typically speaks to the former and rarely to the latter, so our approach restricts the marginal only partially and leaves the completion free; when a baseline prior is substantively meaningful, its conditional distribution can instead be preserved. The resulting bounds make explicit which posterior conclusions follow from the stated economic information and which depend on features of the prior that the researcher cannot defend.

Second, we show that this robustness problem is tractable in set-identified models, including under nonparametric shape restrictions. Conditional on the reduced-form parameter, observationally equivalent structural parameter values have the same likelihood, so the data do not update the conditional prior over the identified set and posterior expectations are linear in it. When the admissible prior information can be expressed through affine restrictions, robust posterior bounds therefore reduce to infinite-dimensional linear programs.

Shape restrictions are not obviously of this form, since they constrain the density of a potentially nonlinear functional rather than the joint prior over which we optimize. Using the Hausdorff moment problem, we characterize weakly decreasing marginal densities and their weak limits through a countable system of moment inequalities that are affine in the joint structural prior, and we extend this characterization to potentially unbounded quantities such as elasticities defined by ratios of impulse responses. Qualitative shape information can thus be imposed alongside conventional probability and moment restrictions, without selecting a parametric density.

Third, we propose a simulation-based implementation and establish its validity. Structural parameter values are drawn from a reference measure, for instance the draws produced by standard sign-restriction samplers, and alternative admissible priors are represented as probability weights on these draws. Robust bounds are then obtained from finite-dimensional linear programs, and we establish conditions under which they converge to their population counterparts, including for posterior probabilities of economic events.

Structural vector autoregressions identified through sign and inequality restrictions provide a natural application. In the traditional sign-restriction approach following Uhlig (2005), economic restrictions act as hard prior information: structural models violating them receive zero probability. Bayesian implementations then place a particular prior over the remaining admissible models, typically the Haar prior on the relevant orthogonal matrices; see, e.g., Arias and Waggoner (2024). Baumeister and Hamilton (2015) instead advocate placing prior distributions directly on economically meaningful parameters such as elasticities. In either case, because the likelihood cannot distinguish observationally equivalent structural models, distributional choices that go beyond the economic information used to justify them permanently influence posterior conclusions.

We revisit the importance of supply and demand shock contributions to economic activity and the real price of oil. Baumeister and Hamilton (2019) challenged the consensus view in the literature, established in Kilian (2009), Kilian and Murphy (2012), and Kilian

and Murphy (2014), that supply shocks are relatively unimportant drivers of the latter two. A central point of disagreement concerns how strongly prior information about short-run oil-market elasticities should restrict the structural model. Starting from the conventional elasticity restrictions in Kilian and Murphy (2014), we allow a controlled prior probability of violating them and additionally consider nonparametric shape restrictions expressing that smaller elasticities are more plausible than larger ones. Varying the probability of violation traces a credibility frontier, similar to the approach in Masten and Poirier (2020), revealing which conclusions survive weaker identifying assumptions. The results separate robust conclusions from assumption-sensitive ones. The combined contribution of demand shocks exceeds that of supply shocks uniformly over prior classes that allow moderate violations of the conventional elasticity restrictions, whereas the stronger claim that supply shocks contribute less than each demand shock individually is not robust. Shape restrictions can substantially sharpen these conclusions. For example, allowing a 10% probability of violating each elasticity restriction, imposing decreasing marginal prior densities narrows the posterior mean bounds on the difference between speculative-demand and supply contributions during the 1990 oil-price spike from $[-55.3, 1.8]$ to $[-49.1, -9.3]$ log points on common feasible reduced-form draws. Thus, qualitative prior information alone can establish a robust ranking.

Beyond the full-ambiguity benchmark, our framework differs from other robust Bayesian approaches in how prior information is specified; Section 3 formalizes the underlying prior decomposition. Giacomini et al. (2019) and Ho (2023) study robustness to departures from a benchmark prior using Kullback–Leibler neighborhoods. Such approaches are natural when the researcher attaches substantive meaning to the baseline prior; ours instead takes the economic restrictions themselves as primitive, without requiring a reference distribution. A different approach is taken by Giacomini et al. (2022), who accommodate uncertainty about identifying assumptions by averaging across alternative models. Probability bounds on violations of identifying restrictions can be represented in this way, but nonparametric shape

restrictions on prior marginals would generally require a continuum of models; our framework imposes them directly. Finally, the marginally specified priors of Kessler et al. (2015) provide a related construction outside the robust Bayesian setting: they replace a baseline marginal with a single prespecified distribution while preserving the baseline conditional distribution. Our framework encompasses this construction as a special case, but also allows the marginal itself to remain partially specified.

Our computational approach also relates to classical robust Bayesian analysis and distributionally robust optimization. Lavine et al. (1991) bound posterior quantities in general Bayesian models with specified prior marginals through a sequence of finite-dimensional linear programs on increasingly refined partitions, an approach that becomes computationally burdensome in high-dimensional parameter spaces. In distributionally robust optimization, Liu et al. (2019) study discrete approximations of moment-constrained ambiguity sets, while Popescu (2005) and Lam et al. (2024) consider optimization under distributional shape restrictions. Our problem differs in both its inferential objective and the structure of the admissible distributions: we use simulated structural draws as support points rather than partitioning the parameter space, and our shape restrictions apply to marginals of nonlinear structural functionals rather than to the distribution over which we optimize.

The remainder of the paper is organized as follows. Section 2 illustrates the main idea in a simple supply–demand example. Section 3 develops the general framework, characterizes the resulting robust posterior bounds, establishes a simulation-based finite-dimensional approximation, and discusses the relation to alternative specifications of prior ambiguity. Section 4 applies the framework to the oil-market SVAR and studies which conclusions about the relative importance of supply and demand shocks remain robust to progressively weaker prior information about short-run elasticities. Section 5 concludes.

2 Illustrative Example

Before outlining the general theory, let us illustrate the main contribution of the paper in a simple supply–demand example. Consider a static market with price p_t and quantity q_t. Supply and demand are

$$q_t = \alpha_s p_t + u_t^s,$$
$$q_t = \alpha_d p_t + u_t^d$$

where α_s is the short-run supply elasticity and α_d is the short-run demand slope. The true values are given by $\alpha_s = 0.25$ and $\alpha_d = -0.50$. The shocks u_t^s and u_t^d shift supply and demand, respectively. The econometrician observes equilibrium prices and quantities. These data can reveal the joint distribution of (p_t, q_t), but they do not in general point-identify the structural slopes (α_s, α_d) without further assumptions. Without knowledge of the true parameters, suppose the econometrician is willing to assume $\alpha_d < 0$ and $\alpha_s > 0$, so demand is downward sloping and supply is upward sloping.

For simplicity, assume the variables are demeaned and let

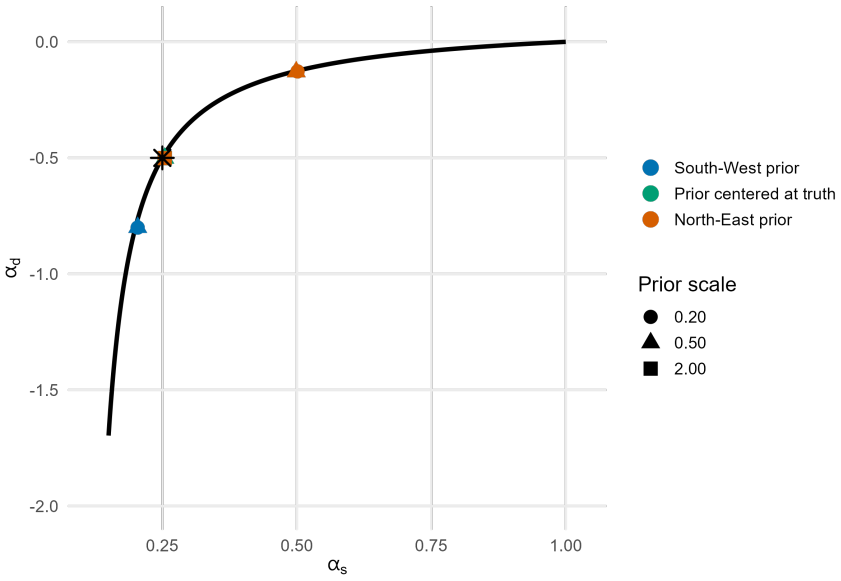
$$\Omega = \begin{pmatrix} \omega_{pp} & \omega_{pq} \\ \omega_{pq} & \omega_{qq} \end{pmatrix}$$

denote the population covariance matrix of equilibrium prices and quantities. Given (α_s, α_d), the structural shocks are $u_t^s = q_t - \alpha_s p_t$ and $u_t^d = q_t - \alpha_d p_t$. Assume that the primitive supply and demand shocks are orthogonal, $\mathrm{Cov}(u_t^s, u_t^d) = 0$. This is the static analogue of assuming a diagonal covariance matrix for structural shocks in SVARs. Orthogonality implies $0 = \omega_{qq} - (\alpha_s + \alpha_d)\omega_{pq} + \alpha_s \alpha_d \omega_{pp}$. Equivalently, for any admissible value of α_s,

$$\alpha_d(\alpha_s) = \frac{\alpha_s \omega_{pq} - \omega_{qq}}{\alpha_s \omega_{pp} - \omega_{pq}}.$$

This curve is the identified set for the two elasticities: every point on it generates the same reduced-form covariance matrix Ω and is therefore observationally equivalent. Conditional on Ω, the likelihood does not choose among these points. Any posterior allocation of mass along the curve must therefore come from prior information. The pure sign restriction $\alpha_s > 0$ leaves a region in which α_s approaches ω_{pq}/ω_{pp} from above and the implied demand elasticity diverges. To keep the illustration focused on prior sensitivity inside a bounded identified set, we impose the maintained lower bound $\alpha_s > 0.15$ throughout. Without this tightening, the same mechanism would be present but visually and numerically more extreme. The identified curve is displayed in Figure 1 as the black solid line.

Figure 1: Identified Set for Elasticities



Suppose now that the researcher has additional, but deliberately soft, prior information about the elasticities. Suppose everyone agrees on the statements:

"Demand is unlikely to be more elastic than -0.5*"*

and

"Supply is unlikely to be more elastic than 0.75*"*

Following the spirit of Baumeister and Hamilton (2015), this information can be encoded by replacing additional hard restrictions with a precise prior distribution over the elasticities. Consider as a baseline a BH-style specification with independent truncated Student-t priors, $\alpha_s \sim t_3(0.50, 0.20^2)$ truncated to $[0.15, \infty)$ and $\alpha_d \sim t_3(-0.125, 0.20^2)$ truncated to $(-\infty, 0)$, where the truncation points correspond to the maintained sign restrictions imposed throughout the example. This is the north-east (NE) prior shown in Figure 1. It provides a natural precise encoding of the two qualitative statements: it assigns only $\Pr(\alpha_s > 0.75) \simeq 0.165$ and $\Pr(\alpha_d < -0.50) \simeq 0.111$ to their respective violations. The difficulty is that doing so necessarily specifies much more than these two economically meaningful probabilities. In particular, it fixes a complete probability distribution over the identified set and thereby favors a particular observationally equivalent structural configuration. With a sample of 5,000 observations, the likelihood is highly informative about the reduced form but essentially uninformative for discriminating among points along the identified set. Consequently, as Figure 1 illustrates, the posterior mode remains close to the NE configuration selected by the prior rather than being pulled toward the true elasticities. This is not a peculiarity of the NE prior: the south-west and truth-centered specifications display the same persistence toward their respective prior locations. Only the prior centered at the truth yields a posterior mode at the truth, but this is tautological: we learn the truth only when we happen to already know it.

A natural response is to conduct conventional prior sensitivity analysis by making the prior increasingly diffuse, as for instance considered in Baumeister and Hamilton (2019). In a point-identified model, this would typically reduce the influence of a badly located prior and allow the likelihood to dominate. Here, however, increasing the common prior scale from 0.20 to 0.50 and 2.00 leaves the posterior mode close to the same observationally equivalent configuration, as shown by the nearly overlapping symbols for each prior location in Figure 1. This result is specific to the simulation choice, however the more general point is that a flat likelihood means that the observed data cannot distinguish among the structural

parameters on the identified curve. It does not mean that these parameters are equally likely. The likelihood measures the relative compatibility of parameter values with the data, it does not assign probabilities to the parameters themselves. A uniform prior instead amounts to assuming equal probability for the values within the identified set, which is quite different from representating ignorance. Thus, sensitivity analysis in set-identified models based on the prior variance are misleading.

In addition, this exercise does not hold fixed the economic information that motivated the prior in the first place. For the NE specification, the violation probabilities increase from $(0.165, 0.111)$ at scale 0.20, to $(0.444, 0.430)$ at scale 0.50, and to $(0.806, 0.825)$ at scale 2.00 for $(\alpha_s > 0.75, \alpha_d < -0.50)$, respectively. Thus, making the prior less precise simultaneously changes the substantive beliefs being imposed: a specification motivated by the view that large supply elasticities and very elastic demand are unlikely eventually assigns most of its probability to precisely those events. Conventional scale sensitivity therefore confounds two conceptually distinct questions: how strongly the researcher wishes to commit to a particular precise prior location and which economic statements the prior is meant to encode.

This suggests a different form of sensitivity analysis: hold fixed the economically meaningful restrictions on these tail probabilities while allowing the otherwise unspecified features of the prior to vary. In particular, consider encoding our beliefs in probability statements:

$$P(\alpha_d \geq -0.50) \geq 1 - \kappa \qquad \text{and} \qquad P(\alpha_s \leq 0.75) \geq 1 - \kappa$$

for different values of κ, i.e. different measures of subjective uncertainty about our prior beliefs.

We propose to bound the posterior quantities of interest over the class of all priors satisfying these probability statements. Importantly, this class retains only the economic content of the original prior information. For example, $\kappa = 0.10$ requires the researcher to assign at least 90% prior probability to demand being no more elastic than -0.50 and at least 90%

prior probability to supply being no more elastic than 0.75, but leaves the remaining shape and allocation of prior probability unrestricted. Varying κ therefore changes the confidence attached to the economic statements themselves rather than perturbing an otherwise arbitrary parametric prior.

The following table reports the resulting bounds on the posterior means of the two elasticities for different values of κ.

Table 1: Credal bounds for Elasticities

Restriction	κ	$E[\alpha_s \mid Y]$	Width α_s	$E[\alpha_d \mid Y]$	Width α_d
Full ambiguity	–	[0.150, 1.004]	100%	[-1.802, -0.000]	100%
Both tail restrictions	0.05	[0.249, 0.762]	60%	[-0.564, -0.038]	29%
Both tail restrictions	0.10	[0.244, 0.775]	62%	[-0.629, -0.036]	33%
Both tail restrictions	0.20	[0.233, 0.800]	66%	[-0.760, -0.032]	40%
Both tail restrictions	0.30	[0.223, 0.826]	71%	[-0.890, -0.028]	48%

Notes: The reported width is the width of the corresponding credal interval relative to the interval obtained under full ambiguity.

Table 1 shows that these weak probability statements can nevertheless be informative. Under full ambiguity, corresponding to the absence of any additional restriction on the prior, the posterior mean of the supply elasticity can range from 0.150 to 1.004, while that of the demand elasticity can range from -1.802 to approximately zero. Imposing only the two tail-probability restrictions substantially contracts these sets. For example, at $\kappa = 0.10$, the admissible posterior means are $[0.244, 0.775]$ for supply and $[-0.629, -0.036]$ for demand, corresponding to only 62% and 33%, respectively, of their full-ambiguity widths.

The comparison also illustrates that weak prior information need not be equally informative about every object. In this example, the same two probability statements discipline the posterior mean of the demand elasticity considerably more strongly than that of the supply elasticity: across the values of κ considered, the demand bounds retain only 29–48% of their full-ambiguity width, compared with 60–71% for supply. Thus, rather than forcing a

precise answer through an arbitrary completion of the prior, the credal analysis reveals which conclusions are actually supported by the combination of the data and the economic beliefs imposed. As κ increases and the statements are weakened, the bounds expand smoothly toward the full-ambiguity benchmark.

The conclusion of this example is not that inference under a single precise prior is inherently inappropriate in set-identified models. If a researcher can defend the full specification of a precise parametric prior, then using that prior is entirely coherent. Our point is that such knowledge is often unavailable. Researchers may instead be willing to defend only weaker statements about the distribution of structural parameters. For example, the belief that smaller values are more plausible than larger values can be represented by the class of weakly decreasing priors; absent further information, we argue that inference should reflect the entire class rather than an arbitrary selection of it. Likewise, as in the preceding example, a researcher may use prior information only to express distrust in a maintained economic restriction, without having a defensible view about the full shape of the prior. In such cases, the likelihood does not resolve uncertainty along the unidentified dimension, so posterior conclusions remain sensitive to how the prior completes the information that the researcher is actually willing to defend. Our proposal is therefore to represent that information directly through classes of distributions and to report the range of posterior conclusions consistent with them.

The preceding example is deliberately simple in another respect: the objects about which prior information is imposed are themselves also the objects of interest. More generally, these need not coincide. Let $S(\theta)$ denote a feature of the structural parameter about which the researcher has credible prior information, while $T(\theta)$ denotes the posterior quantity ultimately of interest. Economic knowledge may then restrict the admissible distribution of $S(\theta)$, which in turn restricts the admissible distributions over the underlying structural parameter θ and thereby the posterior distribution of $T(\theta)$. Schematically, the information

flows as

$$S(\theta) \quad \Longrightarrow \quad \theta \quad \Longrightarrow \quad T(\theta).$$

Thus, the purpose of the restrictions need not be to place direct prior bounds on the target itself which will often shrink ambiguity in a tautological way. They can instead encode credible knowledge about one feature of the model and ask what that knowledge implies for a different economic quantity of interest. The next section formalizes this construction.

3 Theory

We first set up the population posterior sensitivity problem under a class of well-defined prior restrictions. Thereafter, a finite Monte-Carlo approximation to the infinite dimensional linear program is introduced with associated convergence results. As our sensitivity analysis is with respect to the conditional unrevisable part of the prior the next subsection discusses implications for the joint reduced form and structural prior. We conclude with a discussion and relation to other proposals within the literature.

3.1 Set-up and Prior Decomposition

Let $\phi \in \Phi$ denote the reduced-form parameter and let $\theta \in \Theta$ denote the structural parameter, where $(\Phi, \mathcal{A}_\phi)$ and $(\Theta, \mathcal{A}_\theta)$ are standard Borel spaces. The two objects are linked by a measurable many-to-one function $g : \Theta \to \Phi$, $\phi = g(\theta)$, which defines the incompleteness of the structural model and thus the set-identification problem. We assume that the likelihood depends only on ϕ.

Assumption 1. *Denote the observed data by Y and the likelihood function by $p(Y|\cdot)$. Then $p(Y|\phi, \theta) = p(Y|\phi)$.*

Suppose we have a set of assumptions which we are willing to assume with certainty,

denoted by $\mathcal{A}_0$. These could be normalization assumptions as, e.g., in standard SVAR analysis (see for instance the unit effect normalization in Rubio-Ramirez et al. (2010)) or substantive but generally agreed upon assumptions (e.g. impact sign restrictions in the monetary policy example of Uhlig (2005)). Define the identified set as

$$\mathrm{IS}_\theta(\phi) \equiv \{\theta \in \Theta : g(\theta) = \phi, \quad \mathcal{A}_0 \text{ holds}\}.$$

In the absence of further identifying assumptions, this is usually the object of interest for identification analysis in frequentist studies; for reviews see Molinari (2020) and Canay and Shaikh (2017) for inference.

For an ordinary Bayesian approach we need to specify a prior on both parameters which, under standard conditions, can be decomposed as

$$\pi(d\phi, d\theta) = \pi_{\theta|\phi}(d\theta|\phi)\pi_\phi(d\phi).$$

As in Giacomini et al. (2019) and Giacomini and Kitagawa (2021), we focus on sensitivity with respect to the unrevisable component of the prior $\pi_{\theta|\phi}$, as this can never be updated by the data; see e.g. Giacomini and Kitagawa (2021). In general, one could also conduct sensitivity analysis with respect to the reduced-form prior as in Ho (2023).

As Moon and Schorfheide (2012) show, specifying a single precise prior for the structural parameter yields a posterior contracting within the identified set. Thus, from a frequentist point of view, a single-prior Bayesian will be too confident. However, from a Bayesian perspective, as long as we can defend the prior used there is no conceptual problem with using a single prior. The issue is more that, from a substantive point of view, being able to specify an entire joint conditional prior for the structural parameter vector confidently seems implausible. For instance, modern SVAR applications have hundreds of structural parameters. Do we really claim to be able to specify a precise joint law for all of them? Even if so, are we comfortable with all implications such a prior carries for functionals of these parameters?

This paper starts from a different standpoint. Suppose we have knowledge about some implication a full joint conditional prior $\pi_{\theta|\phi}$ should have and leave everything else unspecified. Let $S : \Theta \to \mathbb{R}^{d_s}$ be some transformation of the full parameter vector. Let $\mathcal{M}_S(\phi)$ be the class of admissible distributions of S induced by $\pi_{\theta|\phi}$, i.e. $\mu = S_{\#}\pi_{\theta|\phi}$.

Many joint distributions $\pi_{\theta|\phi}$ can be compatible with a given marginal $\mu \in \mathcal{M}_S(\phi)$. Thus, even if $\mathcal{M}_S(\phi)$ is a singleton, a marginal prior must still be completed into a joint structural prior. To formalize this, for fixed ϕ define the feasible fiber associated with s as

$$F_\phi(s) \equiv \{\theta \in \mathrm{IS}_\theta(\phi) : S(\theta) = s\}.$$

Since the parameter spaces are standard Borel, a joint structural prior can be disintegrated as

$$\pi_{\theta|\phi}(A) = \int \rho(A|s)\mu(ds), \qquad A \in \mathcal{A}_\theta,$$

where $\rho(\cdot|s)$ is a conditional distribution over the structural parameter given $S = s$. Compatibility with the identified set requires $\rho(F_\phi(s)|s) = 1$ for μ-almost every s. We refer to ρ as a completion of the marginal prior μ, and let $\mathcal{R}(\phi)$ denote the class of admissible completion kernels.

Example 1. *Consider a two-dimensional structural parameter $\theta = (\theta_1, \theta_2)$ and suppose that, for a given ϕ, the identified set is*

$$\mathrm{IS}_\theta(\phi) = \{(\theta_1, \theta_2) : 0 \le \theta_1 \le 1,\ 0 \le \theta_2 \le 1 - \theta_1\}.$$

Suppose our prior knowledge concerns only $S(\theta) = \theta_1$, and is precise: $\mathcal{M}_S(\phi) = \{\mu^0\}$ with $\mu^0 = U[0, 1]$. Thus there is no ambiguity about the marginal prior for θ_1. Nevertheless, this does not determine a joint prior for (θ_1, θ_2).

For example, all of the following completion rules are compatible with the same marginal

$\theta_1 \sim U[0,1]$ *and with the identified set:*

$$\rho_1(d\theta_2|s) = \delta_0(d\theta_2),$$
$$\rho_2(d\theta_2|s) = U[0, 1-s](d\theta_2),$$
$$\rho_3(d\theta_2|s) = \delta_{1-s}(d\theta_2).$$

They therefore generate three different joint structural priors despite agreeing exactly on the marginal distribution of S*.*

The distinction is consequential for other structural quantities. For instance, let $T(\theta) = \theta_1 + \theta_2$*. Under the three completions above,*

$$\mathbb{E}_{\pi_1}[T] = \frac{1}{2}, \qquad \mathbb{E}_{\pi_2}[T] = \frac{3}{4}, \qquad \mathbb{E}_{\pi_3}[T] = 1,$$

respectively.

Hence even a singleton class $\mathcal{M}_S(\phi)$ *does not specify a unique structural prior, nor a unique prior for a target of interest. A completion specifies how prior mass is distributed over the remaining dimensions of the identified set conditional on the value of* S*.*

Together, the class of admissible marginals $\mathcal{M}_S(\phi)$ and the completion class $\mathcal{R}(\phi)$ define the set of admissible joint priors for θ conditional on ϕ. For $\mu \in \mathcal{M}_S(\phi)$ and $\rho \in \mathcal{R}(\phi)$, define

$$\pi_{\mu,\rho}(A) \equiv \int \rho(A|s)\mu(ds), \qquad A \in \mathcal{A}_\theta.$$

The corresponding class of admissible structural priors is

$$\Pi_{\mathcal{M},\mathcal{R}}(\phi) \equiv \{\pi_{\mu,\rho} : \mu \in \mathcal{M}_S(\phi),\ \rho \in \mathcal{R}(\phi)\}.$$

The framework nests several existing approaches as special cases.

Example 2. *If both the marginal and the completion are fixed,*

$$\mathcal{M}_S(\phi) = \{\mu^0\}, \qquad \mathcal{R}(\phi) = \{\rho^0\},$$

then $\Pi_{\mathcal{M},\mathcal{R}}(\phi)$ *is a singleton and we recover an ordinary Bayesian analysis with a fully specified conditional structural prior. The commonly used Haar prior in SVAR analysis is an example of such a construction: it specifies a unique joint prior and therefore implicitly fixes both the marginal distribution of any transformation* S *and its completion. See, for example, Arias and Waggoner (2024).*

Example 3. *At the opposite extreme, Giacomini and Kitagawa (2021) leave the unrevisable conditional structural prior unrestricted over the identified set. In our notation, this corresponds to allowing both the marginal and the completion to vary freely:*

$$\mathcal{M}_S(\phi) = \mathcal{P}\left(S(\mathrm{IS}_\theta(\phi))\right), \qquad \mathcal{R}(\phi) = \mathcal{R}^{\text{full}}(\phi) \equiv \{\rho : \rho(F_\phi(s)|s) = 1 \text{ for all feasible } s\}.$$

Consequently, $\Pi_{\mathcal{M},\mathcal{R}}(\phi) = \mathcal{P}\left(\mathrm{IS}_\theta(\phi)\right)$.

Example 4. *For point-identified models Kessler et al. (2015) propose a marginally-specified prior construction. Starting from a baseline joint prior* π^0*, they replace its marginal distribution for a particular functional* S *by a desired marginal* μ^0*, while retaining the baseline conditional distribution given* S*. The resulting joint prior is the KL-closest prior to* π^0 *among all priors having marginal* μ^0*. In our notation, this amounts to fixing*

$$\mathcal{M}_S(\phi) = \{\mu^0\}, \qquad \mathcal{R}(\phi) = \{\rho^0\},$$

where ρ^0 *is the completion inherited from the baseline prior.*

The last example naturally also connects our approach to sensitivity analysis with respect to some baseline prior as in Giacomini et al. (2019) or Ho (2023). We will discuss this further below.

3.2 Population Problem

For some posterior target of interest $T : \Theta \to \mathbb{R}$ a robust Bayes approach given the class $\Pi(\phi)$ aims to find bounds on posterior expectations:

$$\underline{m}_T(\phi) \equiv \inf_{\pi \in \Pi(\phi)} \mathbb{E}_\pi[T \mid Y, \phi] \qquad \bar{m}_T(\phi) \equiv \sup_{\pi \in \Pi(\phi)} \mathbb{E}_\pi[T \mid Y, \phi]\,. \tag{1}$$

This is the inner optimization for a decision problem under ambiguity as defined in Giacomini et al. (2019), which establishes for this case, only the marginal distribution of $T(\theta)$ is relevant for the problem. Nevertheless, as both $S(\theta)$ and $T(\theta)$ depend on the common parameter vector θ, restricting the distribution of $S(\theta)$ can restrict the outcome marginal in a nontrivial way even if $\mathcal{R}(\phi)$ is left unrestricted.

Generically, finding these bounds is a complex task as first $\mathbb{E}_\pi[T \mid Y, \phi]$ is a non-linear function of the prior class elements π. Second, the elements themselves are functions and thus infinite dimensional objects. For the first issue, the defining set-identification property formalized by Assumption 1, which causes the underlying problem with precise priors, also massively simplifies the robust problem. Under Assumption 1:

$$\begin{aligned}
\mathbb{E}_\pi[T \mid Y, \phi] &= \frac{\int T(\theta) p(Y \mid \phi, \theta)\, \pi_{\theta|\phi}(d\theta)}{\int p(Y \mid \phi, \theta)\, \pi_{\theta|\phi}(d\theta)} \\
&= \frac{\int T(\theta) p(Y \mid \phi)\, \pi_{\theta|\phi}(d\theta)}{\int p(Y \mid \phi)\, \pi_{\theta|\phi}(d\theta)} \\
&= \int T(\theta)\, \pi_{\theta|\phi}(d\theta).
\end{aligned}$$

or equivalently $\pi_{\theta|Y,\phi} = \pi_{\theta|\phi}$. In words, conditional on ϕ, posterior expectations with respect to the unrevisable structural prior coincide with prior expectations. Hence the objective in (1) is linear in $\pi_{\theta|\phi}$. If $\Pi(\phi)$ is defined by restrictions linear in $\pi_{\theta|\phi}$, (1) becomes an infinite dimensional linear program.

Proposition 1. *Fix ϕ and suppose $\Pi_{\mathcal{M},\mathcal{R}}(\phi) \neq \emptyset$. Let $T : \Theta \to \mathbb{R}$ be bounded and measurable.*

Then

$$\underline{m}_T(\phi) = \inf_{\pi \in \Pi_{\mathcal{M},\mathcal{R}}(\phi)} \int_\Theta T(\theta)\pi(d\theta) \qquad \bar{m}_T(\phi) = \sup_{\pi \in \Pi_{\mathcal{M},\mathcal{R}}(\phi)} \int_\Theta T(\theta)\pi(d\theta)$$

If in addition $\Pi_{\mathcal{M},\mathcal{R}}(\phi)$ *admits an affine representation, i.e. for a potential infinite index set* J*, measurable functions* a_j *and constants* b_j

$$\Pi_{\mathcal{M},\mathcal{R}}(\phi) = \left\{ \pi \in \mathcal{P}(\mathrm{IS}(\phi)) : \quad \int a_j(\theta)\pi(d\theta) \leq b_j \quad \forall\ j \in J \right\}$$

then $\underline{m}_T(\phi)$ *and* $\bar{m}_T(\phi)$ *are solutions to infinite dimensional linear programs over* π*.*

The proof is given in Appendix A.1.

Proposition 1 is useful because many forms of partial prior knowledge that arise naturally in applications admit an affine representation. We give several examples.

Example 5 (Moment restrictions)**.** *Suppose prior information concerns moments of the economically meaningful transformation* $S(\theta)$*. For measurable functions* $h_j : \mathbb{R}^{d_s} \to \mathbb{R}$*, restrictions of the form*

$$\underline{m}_j \leq \int h_j(S(\theta))\, \pi(d\theta) \leq \overline{m}_j, \qquad j = 1, \ldots, J,$$

are affine in the structural prior π*. Thus, although* h_j *may be nonlinear in the structural parameter, moment restrictions on* $S(\theta)$ *directly satisfy the affine representation in Proposition 1.*

Example 6 (Probability restrictions)**.** *Suppose prior knowledge restricts the probability assigned to some measurable event* $B \subseteq \mathbb{R}^{d_s}$*. Since*

$$P_\pi(S(\theta) \in B) = \int \mathbf{1}\{S(\theta) \in B\}\ \pi(d\theta),$$

restrictions of the form

$$\underline{p} \leq P_\pi(S(\theta) \in B) \leq \overline{p}$$

are affine in π. This includes, for example, bounds on the prior probability that an elasticity or another structural quantity lies above or below a specified threshold.

Example 7 (Shape restrictions on a marginal distribution)**.** *Affine representations can also accommodate nonparametric shape restrictions. Suppose a scalar transformation satisfies $S(\theta) \in [0, B]$ for all $\theta \in \mathrm{IS}_\theta(\phi)$, and prior knowledge requires smaller values of S to be weakly more plausible than larger ones. Let $\mathcal{M}_S(\phi)$ consist of all distributions of the form*

$$\mu(ds) = w_0\, \delta_0(ds) + f(s)\, ds,$$

where $w_0 \in [0, 1]$ and $f : (0, B] \to [0, \infty)$ is weakly decreasing. The atom at zero makes this class weakly closed; it is the weak closure of the class of distributions on $[0, B]$ with a weakly decreasing density. For integers $n \geq 0$ and $k \geq 1$, let

$$q_{n,k}(x) = \frac{d}{dx}\left\{x^{n+1}(1-x)^k\right\}, \qquad x \in [0, 1].$$

Lemma 6 in Appendix B shows that $S_\# \pi \in \mathcal{M}_S(\phi)$ if and only if

$$\int q_{n,k}\big(S(\theta)/B\big)\, \pi(d\theta) \geq 0 \qquad \text{for all } n \geq 0,\ k \geq 1.$$

With an unrestricted completion, $\Pi_{\mathcal{M},\mathcal{R}}(\phi)$ consists of all $\pi \in \mathcal{P}(\mathrm{IS}_\theta(\phi))$ with $S_\# \pi \in \mathcal{M}_S(\phi)$, so it admits the affine representation of Proposition 1 with countably many restrictions. If S is continuous on the identified set, each restriction has a continuous integrand, as required for the finite approximation below. Monotonicity imposed only above a threshold admits an analogous representation; see Appendix B. Lemma 7 in the same appendix extends both

representations to statistics without a known bound, such as ratios whose denominator can vanish on the identified set.

3.3 Finite Approximation

We approximate the infinite-dimensional problem by optimizing over probability measures supported on simulated structural parameters. The probability weights on these points are free decision variables. Under the conditions below, the resulting bounds converge to their population counterparts as the simulated grid grows.

Fix ϕ and write $K_\phi \equiv \mathrm{IS}_\theta(\phi)$. For the approximation result, suppose the population class $\Pi(\phi) \equiv \Pi_{\mathcal{M},\mathcal{R}}(\phi)$ admits the representation

$$\Pi(\phi) = \left\{ \pi \in \mathcal{P}(K_\phi) : \int_{K_\phi} a_j(\theta)\, \pi(d\theta) \le b_j \quad \forall j \in J, \qquad \pi(G_\ell) \le \kappa_\ell \quad \forall \ell \in L \right\}.$$

The first group consists of continuous restrictions. It can include moment restrictions and the shape restrictions characterized in Appendix B. The second group bounds the probabilities of specified events. Separating the two groups is useful because the simulation approximates continuous restrictions and event probabilities in different ways.

When J is infinite, the finite program imposes only the rows in a finite set $J_r \subseteq J$. All event restrictions are imposed at every stage. The corresponding population stage class is

$$\Pi^{(r)}(\phi) = \left\{ \pi \in \mathcal{P}(K_\phi) : \int_{K_\phi} a_j(\theta)\, \pi(d\theta) \le b_j \quad \forall j \in J_r, \qquad \pi(G_\ell) \le \kappa_\ell \quad \forall \ell \in L \right\}.$$

The stage classes must become progressively more restrictive and jointly recover the population class. The row sets J_r themselves need not be nested.

Assumption 2. *(K_ϕ, d) is a nonempty compact metric space.*

Assumption 3. *J is countable and L is finite. Each a_j belongs to $C(K_\phi)$ and $b_j \in \mathbb{R}$. Each G_ℓ is relatively open in K_ϕ, and $\kappa_\ell \in [0,1]$.*

Assumption 4. *For every r, $J_r \subseteq J$ is finite. Moreover, $\Pi^{(r+1)}(\phi) \subseteq \Pi^{(r)}(\phi)$ and $\bigcap_{r\geq 1} \Pi^{(r)}(\phi) = \Pi(\phi)$.*

The approximation theorem covers continuous targets $T \in C(K_\phi)$ and indicator targets $T = \mathbf{1}_A$ for relatively closed $A \subseteq K_\phi$. To describe where the simulated points must lie, form the finite partition $\mathcal{C}$ from the nonempty intersections of each G_ℓ or its complement in K_ϕ. For an indicator target, include A or its complement in these intersections as well. We call the resulting sets *cells*. Membership in every restriction event, and in A when applicable, is constant within a cell.

Fix a simulation measure $\nu \in \mathcal{P}(K_\phi)$ and let $\theta^{(1)}, \theta^{(2)}, \ldots$ be draws from ν. Write $\Theta_N = \{\theta^{(1)}, \ldots, \theta^{(N)}\}$. Every probability measure supported on this grid can be written as $\sum_{n=1}^N w_n \delta_{\theta^{(n)}}$ for some $w \in \Delta_N$. The simulation measure determines which structural points are available; the weights are chosen by the optimization. Its choice can affect the finite-sample bounds and the speed of convergence, even when it does not affect the limiting bounds.

Assumption 5. *The draws $\theta^{(1)}, \theta^{(2)}, \ldots$ are iid with law ν. For every cell $C \in \mathcal{C}$, every $x \in C$, and every $\epsilon > 0$,*

$$\nu\big(B(x,\epsilon) \cap C\big) > 0.$$

Define the sampled stage class by $\Pi_{N,r}(\phi) \equiv \Pi^{(r)}(\phi) \cap \mathcal{P}(\Theta_N)$. The finite approximations to the population bounds in (1) are

$$\underline{m}_{T,N,r}(\phi) \equiv \inf_{\pi \in \Pi_{N,r}(\phi)} \mathbb{E}_\pi[T \mid Y, \phi], \qquad \bar{m}_{T,N,r}(\phi) \equiv \sup_{\pi \in \Pi_{N,r}(\phi)} \mathbb{E}_\pi[T \mid Y, \phi]. \tag{2}$$

By Proposition 1, these objectives equal $\int_{K_\phi} T \, d\pi$. Consequently, both problems are finite linear programs in the weights w_n. The sampled measures satisfy the stage-r restrictions exactly; when J is infinite, they need not satisfy rows omitted from J_r.

The next condition ensures that small errors in continuous restrictions caused by approximating a population prior on the simulated grid can be repaired.

Assumption 6. *For every r, there exists $\pi_r^\circ \in \Pi^{(r)}(\phi)$ such that*

$$\eta_r \equiv \min_{j \in J_r} \left(b_j - \int_{K_\phi} a_j(\theta)\, \pi_r^\circ(d\theta) \right) > 0,$$

with $\eta_r = +\infty$ if $J_r = \varnothing$.

The following condition concerns only the lower bound for an indicator target when the stage classes do not eventually equal $\Pi(\phi)$.

Assumption 7. *For a relatively closed target event $A \subseteq K_\phi$,*

$$\inf_{\pi \in \Pi(\phi)} \pi\big(\mathrm{int}_{K_\phi} A\big) = \inf_{\pi \in \Pi(\phi)} \pi(A).$$

Theorem 1. *Fix ϕ and suppose Assumptions 1- 6 hold. Let $T \in C(K_\phi)$, or let $T = \mathbf{1}_A$ for a relatively closed $A \subseteq K_\phi$. In the indicator case, suppose in addition that either Assumption 7 holds or $\Pi^{(r)}(\phi) = \Pi(\phi)$ for all sufficiently large r. Then there exists a deterministic nondecreasing sequence $r(N) \to \infty$ such that, almost surely,*

$$\underline{m}_{T,N,r(N)}(\phi) \longrightarrow \underline{m}_T(\phi), \qquad \bar{m}_{T,N,r(N)}(\phi) \longrightarrow \bar{m}_T(\phi).$$

The theorem establishes the existence of a suitable schedule $r(N)$; it does not assert convergence for every schedule. If the population class has finitely many continuous rows, a single stage containing all of them suffices. The proof is given in Appendix A.3, and the representation of the shape restrictions is given in Appendix B.

When there are only event restrictions, the result is stronger for an indicator target. In that case no approximation of locations within a cell is needed: the restrictions and objective depend only on the total probability assigned to each cell.

Corollary 1. *Suppose $J = \varnothing$, $\Pi(\phi) \neq \varnothing$, and $T = \mathbf{1}_A$ for a relatively closed $A \subseteq K_\phi$.*

Suppose the draws are iid from ν and $\nu(C) > 0$ for every cell $C \in \mathcal{C}$. Define

$$\mathcal{D} = \left\{ p \in \Delta(\mathcal{C}) : \sum_{C \subseteq G_\ell} p_C \leq \kappa_\ell \quad \forall \ell \in L \right\},$$

where $\Delta(\mathcal{C})$ is the simplex of cell masses. Then

$$\underline{m}_T(\phi) = \min_{p \in \mathcal{D}} \sum_{C \subseteq A} p_C, \qquad \bar{m}_T(\phi) = \max_{p \in \mathcal{D}} \sum_{C \subseteq A} p_C.$$

Almost surely, both sampled bounds equal their population bounds for every sufficiently large N. Equality holds as soon as every nonempty cell contains a simulated point.

The proof is given in Appendix A.4.

Thus, with three event restrictions and one indicator target, the tail-only population problem has at most $2^4 = 16$ cell masses. The corollary requires only positive simulation probability for each cell, a weaker condition than Assumption 5.

3.4 Implied Joint Prior

The previous results are stated conditional on a fixed reduced-form parameter ϕ. They also induce a corresponding class of joint priors and posteriors. Keeping the reduced-form prior π_ϕ fixed, define

$$\Pi^J_{\mathcal{M},\mathcal{R}} \equiv \left\{ \pi(d\phi, d\theta) = \pi_\phi(d\phi)\pi_{\theta|\phi}(d\theta|\phi) : \pi_{\theta|\phi}(\cdot|\phi) \in \Pi_{\mathcal{M},\mathcal{R}}(\phi) \quad \pi_\phi\text{-a.s.} \right\},$$

where $\phi \mapsto \pi_{\theta|\phi}(A|\phi)$ is measurable for every $A \in \mathcal{A}_\theta$. Equivalently, for almost every ϕ,

$$\pi_{\theta|\phi}(A|\phi) = \int \rho_\phi(A|s)\, \mu_\phi(ds), \qquad \mu_\phi \in \mathcal{M}_S(\phi), \quad \rho_\phi \in \mathcal{R}(\phi).$$

All elements of $\Pi^J_{\mathcal{M},\mathcal{R}}$ therefore share the same reduced-form prior, but may differ in both their marginal beliefs about $S(\theta)$ and in how these beliefs are completed into a joint

structural prior. In particular, the implied unconditional distributions of θ and S are

$$\pi_\theta(d\theta) = \int_\Phi \pi_{\theta|\phi}(d\theta|\phi)\,\pi_\phi(d\phi),$$
$$\pi_S(ds) = \int_\Phi \mu_\phi(ds)\,\pi_\phi(d\phi).$$

The joint class is nonempty only if $\Pi_{\mathcal{M},\mathcal{R}}(\phi) \neq \varnothing$ for π_ϕ-almost every ϕ. This is a substantive requirement rather than a technicality. A nonempty identified set does not guarantee a nonempty prior class, because the restrictions defining $\mathcal{M}_S(\phi)$ must be satisfiable by distributions supported on $S(\mathrm{IS}_\theta(\phi))$. For instance, with $\kappa = 0$ the probability restrictions in Section 4 admit no prior if no element of the identified set satisfies all elasticity bounds simultaneously. Similarly, every distribution in the monotone class of Example 7 assigns positive probability to every neighborhood of zero, so this class admits no prior if S is bounded away from zero on the identified set. Whether $\Pi_{\mathcal{M},\mathcal{R}}(\phi)$ is empty depends on ϕ and on the prior restrictions, but not on the data. Following Giacomini and Kitagawa (2021), who trim the reduced-form prior to reduced-form parameters with a nonempty identified set, we restrict π_ϕ to the (assumed measurable) set $\{\phi : \Pi_{\mathcal{M},\mathcal{R}}(\phi) \neq \varnothing\}$ and assume henceforth that $\Pi_{\mathcal{M},\mathcal{R}}(\phi) \neq \varnothing$ for π_ϕ-almost every ϕ. In the application, similar to their report of the posterior probability of a nonempty identified set, we report the number of posterior draws of ϕ for which the prior class is empty.

Under Assumption 1, observing Y updates only the reduced-form marginal. Hence every admissible joint posterior takes the form

$$\pi(d\phi, d\theta|Y) = \pi_\phi(d\phi|Y)\pi_{\theta|\phi}(d\theta|\phi),$$

where $\pi_\phi(\cdot|Y)$ is common across all elements of the prior class and $\pi_{\theta|Y,\phi} = \pi_{\theta|\phi}$.

The joint class is rectangular across reduced-form values: apart from measurability, an admissible element of $\Pi_{\mathcal{M},\mathcal{R}}(\phi)$ can be selected separately for each ϕ. Hence, under standard

measurable-selection conditions,

$$\underline{m}_T(Y) = \int_\Phi \underline{m}_T(\phi)\, \pi_\phi(d\phi|Y), \qquad \overline{m}_T(Y) = \int_\Phi \overline{m}_T(\phi)\, \pi_\phi(d\phi|Y).$$

Thus the conditional optimization problems studied above aggregate directly over the posterior distribution of the reduced form.

Although these posterior bounds aggregate pointwise in ϕ, it is also useful to understand the unconditional implications of restrictions imposed on the conditional marginal distribution of S. The following result shows that two restrictions used below are preserved under mixing over the reduced form. Importantly, these conclusions depend only on the marginal class $\mathcal{M}_S(\phi)$ and therefore hold for any admissible completion class $\mathcal{R}(\phi)$.

Proposition 2. *Let $\pi \in \Pi^J_{\mathcal{M},\mathcal{R}}$ and denote $\mu_\phi = S_\# \pi_{\theta|\phi}$.*

1. *Suppose that for some measurable set $B \subseteq \mathbb{R}$ and $\kappa \in [0,1]$, $\mu_\phi(B) \leq \kappa$ for π_ϕ-almost every ϕ. Then $\pi(S(\theta) \in B) \leq \kappa$.*

2. *Fix $c \in \mathbb{R}$ and let $Z = |S - c|$. Suppose that, for π_ϕ-almost every ϕ, the conditional distribution of Z is $w(\phi)\delta_0(dz) + f_{Z|\phi}(z)\, dz$, where $w(\phi) \in [0,1]$ and $f_{Z|\phi}$ is supported on $[0, \bar{z}(\phi)]$ and non-increasing on $(0, \bar{z}(\phi)]$. Then the unconditional distribution of Z is $\bar{w}\, \delta_0(dz) + f_Z(z)\, dz$ with $\bar{w} = \int_\Phi w(\phi)\, \pi_\phi(d\phi)$ and*

$$f_Z(z) = \int_\Phi f_{Z|\phi}(z)\, \pi_\phi(d\phi),$$

where $f_{Z|\phi}(z) = 0$ for $z > \bar{z}(\phi)$ and each $f_{Z|\phi}$ is taken right-continuous on $(0,\infty)$, and f_Z is non-increasing on $(0,\infty)$.

The same conclusions hold for the corresponding posterior distributions after replacing π_ϕ by $\pi_\phi(\cdot|Y)$.

The proof is given in Appendix A.5.

In general, conditional restriction will not survive the aggregation, however, the type of restrictions in the preceding result are important, as they nest for instance the motivating example of modeling partial trust in some assumption as well as monotone violation from fixed bounds if the assumption consists of inequality constraints.

Sometimes conditional restrictions are more natural than unconditional ones. Consider for instance a missing data problem for which we observe the distribution of a target variable for some share of the population and not for the other one. Conditional on the realized distribution we might want to change our prior beliefs about the missing distribution.

3.5 Sensitivity with Partial Trust in Reference Measure

In the baseline-free specification above, the reference measure used for simulation serves only a computational role and, subject to the conditions of Theorem 1, does not affect the population robust bounds. This differs from approaches to Bayesian robustness that measure departures from a particular baseline prior, as in the KL-based sensitivity analysis of Giacomini et al. (2019). At the same time, it is related to the marginally specified prior construction of Kessler et al. (2015), which modifies a specified marginal distribution while preserving the conditional structure of a baseline prior.

In some applications, however, the researcher may attach substantive meaning to a baseline conditional prior $\pi^0_{\theta|\phi}$ without being willing to accept all of its marginal implications for S. The substantive baseline need not coincide with the measure used for simulation. Let $\pi^0_{S|\phi} \equiv S_{\#}\pi^0_{\theta|\phi}$ and denote the baseline completion by ρ^0_ϕ, so that

$$\pi^0_{\theta|\phi}(A) = \int \rho^0_\phi(A \mid s)\, \pi^0_{S|\phi}(ds), \qquad A \in \mathcal{A}_\theta.$$

We consider the specialization $\mathcal{R}(\phi) = \{\rho^0_\phi\}$, while allowing $\mu \in \mathcal{M}_S(\phi)$. Thus ambiguity remains about the marginal distribution of S, but its completion is inherited from the baseline. Under absolute continuity, this is equivalent to reweighting the baseline by a function

of S alone.

Proposition 3. *Fix $\phi \in \Phi$ and let $\pi^0_{\theta|\phi}$ be a baseline conditional prior supported on K_ϕ, with induced marginal $\pi^0_{S|\phi}$ and completion ρ^0_ϕ. Suppose*

$$\emptyset \neq \mathcal{M}_S(\phi) \subseteq \left\{\mu \in \mathcal{P}(\mathbb{R}^{d_s}) : \mu \ll \pi^0_{S|\phi}\right\},$$

and set $\mathcal{R}(\phi) = \{\rho^0_\phi\}$. For every $\mu \in \mathcal{M}_S(\phi)$, the unique prior $\pi^\mu_{\theta|\phi} \in \Pi_{\mathcal{M},\mathcal{R}}(\phi)$ with $S_\# \pi^\mu_{\theta|\phi} = \mu$ satisfies

$$\begin{aligned}
\pi^\mu_{\theta|\phi}(A) &= \int \rho^0_\phi(A \mid s)\, \mu(ds), \qquad A \in \mathcal{A}_\theta, \\
\pi^\mu_{\theta|\phi}(d\theta) &= \frac{d\mu}{d\pi^0_{S|\phi}}(S(\theta))\, \pi^0_{\theta|\phi}(d\theta).
\end{aligned}$$

Moreover,

$$\mathrm{KL}\left(\pi^\mu_{\theta|\phi} \,\|\, \pi^0_{\theta|\phi}\right) = \mathrm{KL}\left(\mu \,\|\, \pi^0_{S|\phi}\right).$$

If this divergence is finite, $\pi^\mu_{\theta|\phi}$ is the unique solution to

$$\min_{\substack{q_{\theta|\phi} \ll \pi^0_{\theta|\phi} \\ S_\# q_{\theta|\phi} = \mu}} \mathrm{KL}\left(q_{\theta|\phi} \,\|\, \pi^0_{\theta|\phi}\right).$$

The proof is given in Appendix A.6.

Absolute continuity ensures that the construction does not depend on the version of ρ^0_ϕ chosen on $\pi^0_{S|\phi}$-null sets. This is the domination condition imposed by Kessler et al. (2015). If $\mathcal{M}_S(\phi) = \{\mu^\star\}$, Proposition 3 reduces to their marginally specified prior construction. Here, an entire class $\mathcal{M}_S(\phi)$ generates a corresponding class of structural priors, each retaining the same baseline completion. The KL chain rule explains the minimizing property: once the marginal is fixed, changing the completion adds a nonnegative conditional divergence.

Proposition 3 also clarifies the connection to Giacomini et al. (2019). Let

$$\bar{\lambda}(\phi) = \sup_{\mu \in \mathcal{M}_S(\phi)} \mathrm{KL}\left(\mu \,\|\, \pi^0_{S|\phi}\right).$$

If $\bar{\lambda}(\phi) < \infty$, every prior in $\Pi_{\mathcal{M},\{\rho^0_\phi\}}(\phi)$ belongs to the structural KL ball of radius $\bar{\lambda}(\phi)$ around $\pi^0_{\theta|\phi}$. The converse need not hold: a structural KL ball also permits changes to the completion, whereas our construction fixes it. If $\mathcal{M}_S(\phi)$ is itself a KL ball of finite radius λ around $\pi^0_{S|\phi}$, then for bounded losses depending on θ only through S, the worst-case expected loss coincides with that over the structural KL ball of radius λ. This is the marginal-reduction result of Lemma 2.1 in Giacomini et al. (2019). For targets T not determined by S, the two robust analyses need not coincide, because fixing the completion then restricts decision-relevant prior variation.

3.6 Full Ambiguity versus Partially Specified Priors

The distinction between our approach and the full ambiguity robust approach of Giacomini and Kitagawa (2021) can be best understood within the simple example proposed by Moon and Schorfheide (2012). This also provides an excellent abstract motivation for the current paper. Let $y_i = \phi + \varepsilon_i$ with $\varepsilon_i \sim N(0,1)$ so the data identify ϕ. Suppose we know with certainty that the structural parameter θ satisfies $\phi \leq \theta \leq \phi + \lambda$ for known $\lambda > 0$. The sharp identified set, which is the main object of interest from a frequentist perspective, is $\mathrm{IS}_\theta(\phi) = [\phi, \phi + \lambda]$.

A single-prior Bayesian specifies one conditional prior $\pi_{\theta|\phi}$ instead. Asymptotically, prior values outside $\mathrm{IS}_\theta(\phi)$ are assigned zero mass, whereas the posterior inside the identified set is just proportional to the prior, see Moon and Schorfheide (2012) or Baumeister and Hamilton (2015). This is because the likelihood depends only on ϕ, thus this conditional prior is not updated by the data. For an in-depth discussion of whether the shape of the posterior inside the identified set can still be useful see Gustafson (2014). A full ambiguity-robust

Bayesian in the sense of Giacomini and Kitagawa (2021) instead allows every conditional prior supported on the identified set. In the notation above, this corresponds to

$$\mathcal{M}_S(\phi) = \mathcal{P}(S(\mathrm{IS}_\theta(\phi))), \qquad \mathcal{R}(\phi) = \mathcal{R}^{\text{full}}(\phi),$$

and hence $\Pi_{\mathcal{M},\mathcal{R}}(\phi) = \mathcal{P}([\phi, \phi + \lambda])$.

Giacomini and Kitagawa (2021) show that the corresponding set of posterior means coincides with the identified set. In particular, this implies that there exists $\pi^*_{\theta|\phi} \in \Pi_{\mathcal{M},\mathcal{R}}(\phi)$ such that

$$\mathbb{E}_{\pi^*}[\theta \mid y, \phi] = \phi + \lambda.$$

This is only possible if $\pi^*_{\theta|\phi} = \delta_{\phi+\lambda}$. Such priors are formally admissible under full ambiguity, but may be substantively implausible if the researcher has additional prior knowledge.

As above, we might have such additional knowledge in terms of qualitative restrictions. For illustration, write $\theta = \phi + \lambda U$ with $U \in [0, 1]$ and restrict the density of U to be weakly decreasing:

$$\Pi^{\downarrow}_{\theta|\phi} = \left\{ \pi_{\theta|\phi} : \theta = \phi + \lambda U,\ P_U \ll \text{Leb},\ \frac{dP_U}{du} \text{ is weakly decreasing on } [0,1] \right\}.$$

This simply imposes a qualitative shape restriction on the conditional prior for the scalar structural parameter: values of θ closer to the lower endpoint ϕ are weakly more likely than values closer to $\phi + \lambda$. Then the closure of the set of posterior means is

$$\left[\phi, \phi + \frac{\lambda}{2}\right].$$

A proof is provided in Appendix A.7. Thus, a qualitative shape restriction reduces ambiguity without forcing a fully specified prior distribution. Note that, as here the parameter is univariate there is no completion ambiguity.

In this case, the set of posteriors is shrunk purely by assumption. However, these qualitative shape restrictions can be motivated from a Bayesian perspective. For instance, assume θ captures the distance from the actual policy rate to the optimal one. A decision maker may regard small adjustments to the actual rate as ex ante more plausible or may use such a class as a regularizing ambiguity set around the benchmark. A monotone shape restriction as above captures this and can be seen as a soft tie-breaking rule. Independent of such a justification, the example should only be seen as illustrative. In richer models, $S(\theta)$ need not determine θ, so restrictions on $\mathcal{M}_S(\phi)$ interact with the completion class $\mathcal{R}(\phi)$ in determining the admissible distribution of the target $T(\theta)$.

4 Robust Conclusions in the Oil Market

The relative importance of supply and demand disturbances in driving oil-price fluctuations has been debated for almost two decades. The influential analysis of Kilian (2009) attributes much of the variation in real oil prices to demand-side disturbances rather than exogenous disruptions to oil production. Subsequent work by Kilian and Murphy (2012) and Kilian and Murphy (2014) reaches broadly similar conclusions while replacing recursive identification with sign restrictions and additional economically motivated bounds. A particularly debated restriction is that the within-month price elasticity of oil supply cannot exceed 0.0258.

The treatment of this elasticity has subsequently become one of the central points of disagreement in the literature. Baumeister and Hamilton (2019) argue that interpreting 0.0258 as a hard identifying restriction treats uncertain economic information as known with certainty. They instead place explicit prior distributions on economically meaningful structural parameters and obtain a larger role for supply disturbances. Caldara et al. (2019), using external information on oil-market elasticities, similarly emphasize that relatively small changes in admissible supply and demand elasticities can materially change the decomposi-

tion of oil-price fluctuations. In contrast, Kilian (2022) argues that the evidence continues to support a very small within-month supply elasticity and that allowing too much prior mass on large elasticities removes important identifying information.

The disagreement is therefore not simply about whether prior information should be used. Rather, it concerns which features of that information support particular economic conclusions. This is precisely the setting for the framework developed above. Instead of selecting one fully specified structural prior, we ask which conclusions hold uniformly over classes of priors encoding economically interpretable restrictions on elasticity distributions. The application distinguishes the information needed to establish aggregate demand dominance from that needed to determine individual shocks' quantitative contributions.

We use the four-variable monthly oil-market VAR of Kilian and Murphy (2014), containing global oil production, global real activity, the real price of oil, and changes in oil inventories. The maintained identifying assumptions, $\mathcal{A}_0$, consist of impact and dynamic sign restrictions distinguishing an oil supply shock, a flow-demand shock, a speculative-demand shock, and a residual shock. Across the comparisons below, we hold the data, VAR specification, reduced-form prior, shock normalizations, and maintained sign restrictions fixed, and vary the information imposed on the short-run elasticities.

Note that the literature has extended the workhorse model in several directions. One strand adds narrative information. Antolin-Diaz and Rubio-Ramirez (2018) supplement the sign and elasticity restrictions of Kilian and Murphy (2012) with narrative restrictions requiring that adverse oil supply shocks were the main driver of unexpected production changes in months of well-documented disruptions, such as the Iraqi invasion of Kuwait in August 1990. Zhou (2020) imposes narrative sign restrictions in the Kilian and Murphy (2014) model and finds its main conclusions largely unchanged, although whether such restrictions were already part of the original analysis is itself disputed (Kilian and Zhou, 2019; Baumeister and Hamilton, 2020; Kilian, 2022). Read and Zhu (2025) apply the robust Bayesian approach of Giacomini and Kitagawa (2021) to the model of Antolin-Diaz and Rubio-Ramirez (2018) to

assess the influence of the conditional prior and of individual narrative restrictions, and Gazzani et al. (2024) combine proxies with sign and narrative restrictions to identify oil shocks at daily frequency. Other work changes the source of identification or the level of aggregation, exploiting high-frequency variation around OPEC announcements (Känzig, 2021), strategic interactions between OPEC and non-OPEC producers (Alonso-Alvarez et al., 2022), or heterogeneity in supply and demand elasticities across producing and consuming countries (Baumeister and Hamilton, 2023). Herrera and Rangaraju (2020) find that much of the dispersion in estimated effects of oil supply shocks across studies stems from differences in identifying assumptions, notably the admissible short-run supply elasticity. Our goal is not to propose a new oil-market model, discuss the relevant data or to add identifying information, but to explain the original supply and demand debate via looking at prior features. We therefore deliberately stay close to the original specification of Kilian and Murphy (2014).

For every admissible structural model, we compute two measures of the short-run supply elasticity. Let α_s^F denote the impact response of oil production relative to the impact response of the oil price following a flow-demand shock, and let α_s^S denote the analogous ratio following a speculative-demand shock. Both are nonnegative under the maintained sign restrictions. We also compute the impact price elasticity of demand in use, ϵ_d, from the response of oil use, measured as production net of inventory accumulation, to an oil supply shock.

The first ingredient of the prior classes replaces the hard elasticity restrictions by

$$P_\pi\big(\alpha_s^F \leq 0.0258 \mid \phi\big) \geq 1 - \kappa_s, \tag{3}$$

$$P_\pi\big(\alpha_s^S \leq 0.0258 \mid \phi\big) \geq 1 - \kappa_s, \tag{4}$$

$$P_\pi(-0.8 \leq \epsilon_d \leq 0 \mid \phi) \geq 1 - \kappa_d. \tag{5}$$

Thus, each supply bound may be violated with probability at most κ_s, while the union event $\{\epsilon_d < -0.8\} \cup \{\epsilon_d > 0\}$ has probability at most κ_d. The two supply restrictions remain separate probability inequalities. At $\kappa = (\kappa_s, \kappa_d) = (0, 0)$, they become hard support restric-

tions; positive values relax the confidence attached to those restrictions without selecting a distribution within either the admissible or violation regions.

The *tail-only* class imposes only (3)–(5), in addition to $\mathcal{A}_0$. The *global monotonicity* class also restricts the conditional marginal distributions of α_s^F, α_s^S, and $|\epsilon_d|$ to have weakly decreasing densities, in the sense of Example 7. This formalizes the preference for small elasticity magnitudes discussed by Kilian (2022), without choosing a parametric density or fixing its scale. Monotonicity is substantive distributional information, distinct from an upper bound: it disciplines the allocation of probability within the admissible range as well as in the tails. The remaining features of the conditional structural prior are unrestricted.

This distinction is the methodological focus of the application. Relative to full ambiguity over distributions on an identified set, the framework allows the researcher to retain selected distributional implications of economic knowledge. It also complements the treatment of uncertain identifying assumptions in Giacomini et al. (2022): here uncertainty is represented directly through restrictions on conditional structural distributions, and probability bounds can be combined with nonparametric shape information. The main results use the degree-6 Hausdorff approximation; degrees 4 and 8 provide sensitivity checks. Appendix C describes the construction and its relation to Theorem 1.

Conditional bounds are obtained by optimizing probability weights on simulated admissible structural draws and then averaged over reduced-form posterior draws. To isolate the information supplied by monotonicity, tail-only and shape-restricted endpoints are compared on common feasible reduced-form draws within each specification. Thus, each comparison holds the reduced-form distribution fixed while changing the admissible conditional structural prior. We denote the relevant common set by F; the probabilities and expectations below condition on $\phi \in F$. The historical table also reports tail-only bounds on that class's own feasible draws. These two summaries distinguish the conclusions of each maintained specification from the incremental identifying contribution of shape information.

4.1 The Importance of Oil Demand versus Supply Shocks

We first ask whether the ranking of demand and supply survives variation over the admissible prior class. Let $V^s_{p,h}(\theta)$, $V^f_{p,h}(\theta)$, and $V^{sp}_{p,h}(\theta)$ denote the h-month forecast-error variance shares of the real oil price attributable to supply, flow-demand, and speculative-demand shocks. Define the combined demand contribution as $V^d_{p,h}(\theta) = V^f_{p,h}(\theta) + V^{sp}_{p,h}(\theta)$. The denominator includes all four structural shocks; the residual shock is not included in this demand aggregate.

The left panels of Figure 2 consider

$$A_D = \left\{ V^d_{p,12}(\theta) \geq V^s_{p,12}(\theta) \right\}, \tag{6}$$

the event that the two named demand shocks jointly explain at least as much 12-month forecast-error variance as supply. For each prior class $c \in \{T, G\}$, denoting tail-only and global monotonicity, respectively, the reported bounds are

$$\underline{p}^c_D(\kappa; F) = \inf_{\pi \in \Pi^c_\kappa} P_\pi(A_D \mid Y, \phi \in F), \qquad \overline{p}^c_D(\kappa; F) = \sup_{\pi \in \Pi^c_\kappa} P_\pi(A_D \mid Y, \phi \in F). \tag{7}$$

The lower endpoint is the posterior probability under the admissible prior least favorable to demand dominance. Unlike a probability calculated under one selected prior, it quantifies the confidence in this conclusion warranted uniformly over the class.

The figure shows strong support for aggregate demand dominance when the supply-elasticity restrictions are sufficiently credible. For $\kappa_d = 0$, the lower probability at $\kappa_s = 0$ is approximately 0.92 under tail restrictions alone and 0.96 with global monotonicity. At $\kappa_s = 0.1$, the corresponding values are about 0.75 and 0.80. Allowing violations therefore weakens, but does not immediately overturn, the conclusion. At $\kappa_s = 0.3$, the lower probabilities fall to approximately 0.43 and 0.48, so neither class establishes that demand dominance is more likely than not. The upper probabilities remain close to one. Raising κ_d to 0.3 changes these

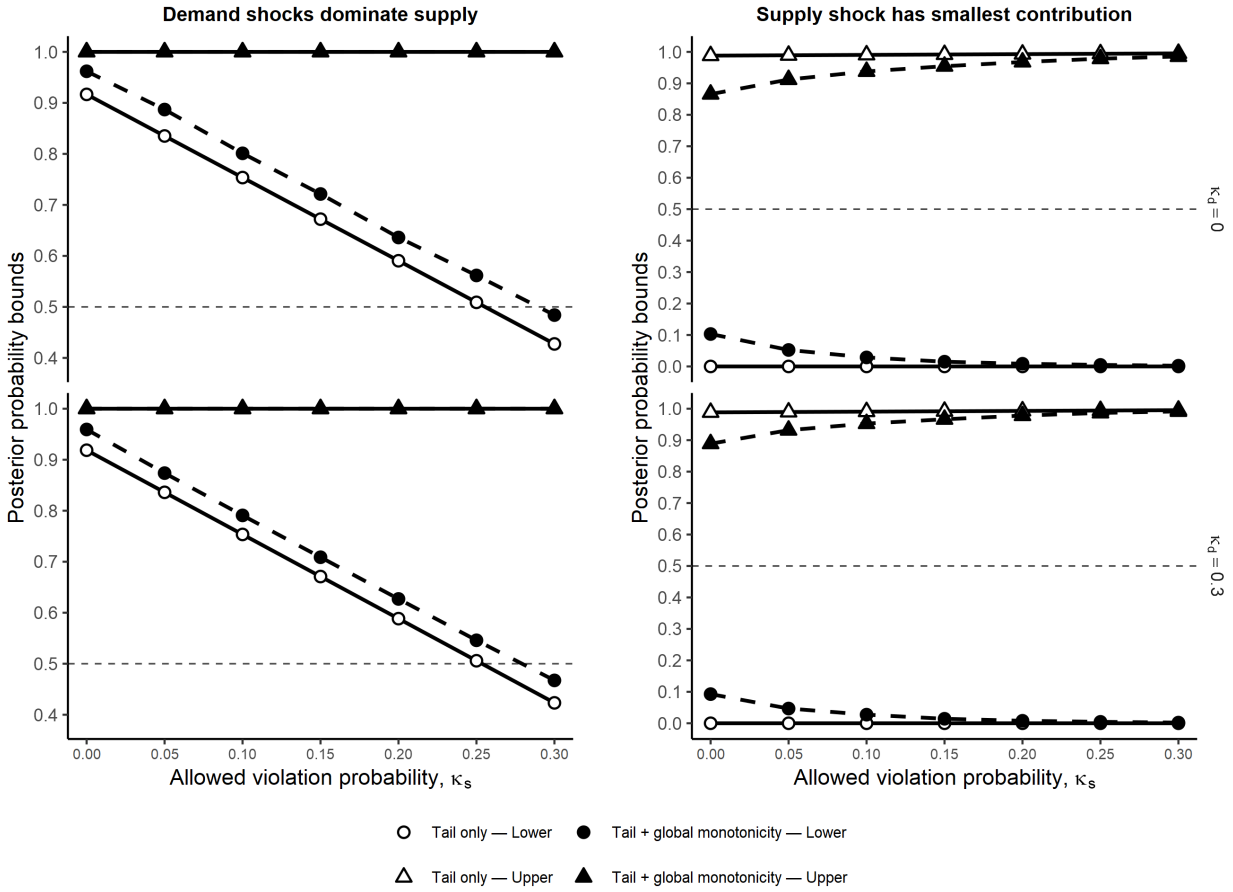
results little. Within each specification, monotonicity raises the lower probability by roughly four to five percentage points.

The right panels ask a more demanding question:

$$A_S = \left\{ V^s_{p,12}(\theta) \leq \min\left\{ V^f_{p,12}(\theta), V^{sp}_{p,12}(\theta) \right\} \right\}.$$

This is the event that supply contributes no more than either named demand shock individually, not that its contribution is economically negligible. The tail-only bounds are essentially $[0, 1]$ on common draws even under hard elasticity restrictions. At $(\kappa_s, \kappa_d) = (0, 0)$, global monotonicity narrows the interval to approximately $[0.10, 0.87]$, but does not establish either ranking with probability exceeding one half under every admissible prior. As the restrictions are relaxed, this interval also approaches $[0, 1]$.

Figure 2: Robust posterior probabilities of oil-price variance rankings



Notes: Bounds on posterior probabilities of A_D (left) and A_S (right) at the 12-month horizon. Tail-only and degree-6 global-monotonicity bounds use common feasible reduced-form draws for the relevant comparison. The horizontal reference line is 0.5. The supply ranking excludes the residual shock. Common feasible sets can differ across specifications; comparisons between prior classes at a given specification hold the reduced-form draws fixed.

The distinction between these events is economically consequential. Elasticity information can support the collective importance of demand without determining the ordering of individual shocks. Nor does a high probability of A_D imply that supply explains a small absolute share of price variation. We therefore turn from probabilities of rankings to the magnitudes of the variance contributions.

Figure 3 reports bounds on

$$\mathbb{E}_\pi\left[V^r_{p,12}(\theta) \mid Y, \phi \in F\right], \qquad r \in \{s, f, sp\},$$

over each admissible prior class. Each horizontal segment is a range of posterior expectations, not a credible interval for a variance share. Endpoints for different shocks are optimized separately and need not be attained by the same prior.

The solid intervals show that hard elasticity restrictions identify an important ranking but leave substantial ambiguity about magnitudes. At $\kappa = (0, 0)$, the tail-only expected supply share ranges approximately from 4% to 28%, while the flow-demand share ranges from 39% to 87%. The separation of these intervals establishes that flow demand contributes more than supply in posterior expectation under every prior in this class. It does not, however, pin down a small supply contribution: values above one quarter remain admissible. The speculative-demand share is less tightly determined, ranging from approximately 2% to 46%. These marginal ranges are consistent with the uncertainty about individual rankings in Figure 2, although the ranking probabilities cannot be recovered from marginal expectation bounds alone.

Shape information is substantially more consequential for these quantitative targets. Under hard restrictions, monotonicity narrows the expected supply share to approximately 7–21%, the flow-demand share to 49–79%, and the speculative-demand share to 6–33%. It removes roughly two fifths of the remaining interval width, despite all prior mass already satisfying the conventional elasticity bounds. Tightening remains visible at positive violation allowances, although its relative magnitude is smaller. Thus, information about how probability is distributed across admissible structural models can materially sharpen quantitative attribution even when it adds relatively little to the probability of an aggregate ranking.

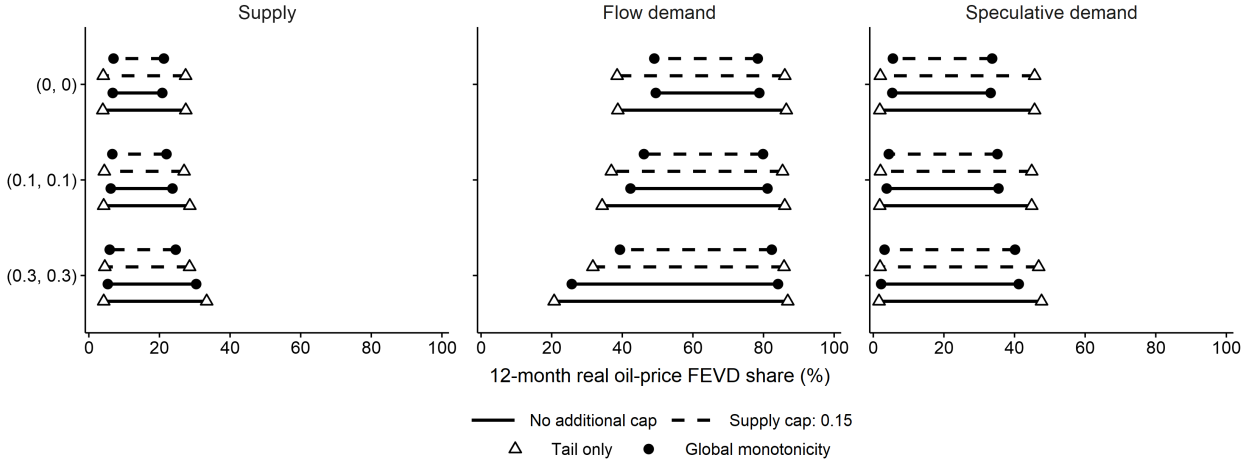
Kilian (2022) suggest that the imposed support for the supply elasticities is more consequential than the specific prior chosen on it. To distinguish this distributional information

from an absolute support restriction, we repeat the exercise imposing

$$P_\pi\big(\alpha_s^F \le 0.15,\ \alpha_s^S \le 0.15 \mid \phi\big) = 1, \tag{8}$$

while retaining the original probability bounds at 0.0258 and the demand-elasticity restrictions. The dashed intervals report this experiment. The ceiling is deliberately generous relative to the conventional supply bound. It excludes elasticity values above 0.15 without requiring the original 0.0258 restriction to hold almost surely when $\kappa_s > 0$. At $\kappa_s = 0$, the additional ceiling is redundant.

Figure 3: Robust expected variance contributions: elasticity support and prior shape



Notes: Ranges of posterior expected 12-month real-oil-price FEVD shares, expressed in percent. Solid intervals impose no additional supply ceiling; dashed intervals additionally impose (8). Open triangles denote tail-only bounds and filled circles denote bounds with global monotonicity at degree 6. Within each support specification and each (κ_s, κ_d), tail-only and shape-restricted intervals use common feasible reduced-form draws. Shape restrictions apply to the two supply-elasticity marginals and the absolute demand-elasticity marginal. The residual shock enters the FEVD denominator but is not displayed.

The capped results show that the identifying value of shape information is not confined to disciplining arbitrarily large supply elasticities. At $\kappa = (0.3, 0.3)$, for example, the capped tail-only supply interval is approximately 4–28%, and monotonicity narrows it to about 6–25%. For flow demand, the corresponding lower endpoint rises from roughly 32% to 39%. The within-support comparisons therefore continue to reveal information in distributional

shape after large supply elasticities have been excluded. This does not establish that shape is universally more important than support, or that a tighter ceiling would be inconsequential. It establishes the narrower and economically relevant point that the ceiling alone does not exhaust the identifying content of partial prior knowledge. Moreover, the shape class restricts both supply-elasticity marginals and the absolute demand-elasticity marginal; its contribution is the joint effect of this distributional information.

These findings distinguish two aspects of the oil-market debate. The conventional elasticity restrictions can support demand dominance without a precise conditional structural prior, while allowing substantial variation in individual shocks' expected contributions. The additional shape restrictions quantify how much of that ambiguity can be removed by qualitative distributional information, rather than by committing to a complete parametric prior.

4.2 Implications for Historical Decompositions

We next ask whether these distinctions matter for the attribution of observed oil-price movements. Let $H_E^r(\theta)$ denote the cumulative contribution of shock r to the change in the log real oil price over episode E, recovered from the structural moving-average representation with a truncation lag of 100 months. Table 2 considers three targets: H_E^s for January 2007–June 2008; $H_E^f - H_E^s$ for January 2003–June 2008; and $H_E^{sp} - H_E^s$ for July–October 1990. Contrasts are optimized directly, rather than formed by subtracting separately optimized endpoints. The table reports ranges of posterior expectations under the baseline tail and degree-6 shape classes, without the additional 0.15 ceiling.

The historical results reinforce the distinction between relative and absolute importance. On the tail-only class's own feasible draws, the expected flow-demand-minus-supply contribution to the 2003–2008 boom lies in $[39.9, 100.3]$ log-price percentage points under hard restrictions and $[22.4, 108.1]$ when each restriction may be violated with probability 0.1. Flow demand therefore contributed more than supply in posterior expectation throughout these prior classes. At violation probabilities of 0.3, the interval becomes $[-12.7, 123.7]$, and this

conclusion is no longer uniform. Nevertheless, supply can make an economically substantial contribution during a particular episode: at $\kappa = (0.1, 0.1)$, the shape-restricted upper expected supply contribution to the 2007–2008 increase is 26.3 log points, approximately 34% of the observed 76.6-log-point increase. Demand dominance over the longer boom does not imply negligible supply effects during the shorter run-up.

Table 2: Robust posterior expectations of historical oil-price contributions

		Tail only	Common draws			Global monotonicity
κ_s	κ_d	Own draws	Tail	Global	Width cut (%)	Share
Supply contribution, January 2007–June 2008 ($\Delta p = 76.6$)						
0	0	[2.4, 26.1]	[−5.1, 30.5]	[2.1, 23.0]	41.4	0.28
0.1	0.1	[0.0, 29.0]	[−6.1, 32.1]	[0.0, 26.3]	31.4	0.42
0.3	0.3	[−4.9, 34.9]	[−6.3, 36.5]	[−2.4, 33.7]	15.8	0.59
Flow demand minus supply, January 2003–June 2008 ($\Delta p = 137.7$)						
0	0	[39.9, 100.3]	[27.0, 118.8]	[48.6, 104.4]	39.3	0.28
0.1	0.1	[22.4, 108.1]	[11.1, 117.7]	[27.5, 105.3]	27.0	0.42
0.3	0.3	[−12.7, 123.7]	[−15.3, 129.4]	[−5.2, 119.9]	13.5	0.59
Speculative demand minus supply, July–October 1990 ($\Delta p = 74.8$)						
0	0	[−51.0, −16.3]	[−55.0, 1.1]	[−46.6, −13.1]	40.3	0.28
0.1	0.1	[−53.6, −9.6]	[−55.3, 1.8]	[−49.1, −9.3]	30.5	0.42
0.3	0.3	[−58.7, 3.8]	[−58.8, 8.0]	[−55.4, 0.2]	16.8	0.59

Notes: Bounds on posterior expected historical contributions and contrasts, in log-price percentage points. They are not credible intervals for historical contributions. Own-draw intervals average conditional endpoints over reduced forms for which the tail-only class is feasible; common-draw intervals use the intersection of tail-only and degree-6 shape-feasible draws at the same (κ_s, κ_d). Width cut is $100(1 - w_G/w_T)$, with both widths calculated on common draws. Share is the fraction of the 300 reduced-form draws admitting the global shape class. Each reduced-form draw uses 200 million proposed structural rotations. No additional 0.15 supply ceiling is imposed.

The common-draw comparisons isolate substantial gains from shape information. Across

the three historical targets, monotonicity reduces interval widths by approximately 40% under hard restrictions and 27–31% at $\kappa = (0.1, 0.1)$. For the latter specification, the expected flow-demand-minus-supply interval narrows from $[11.1, 117.7]$ to $[27.5, 105.3]$. The 1990 episode also illustrates a change in a sign conclusion: on common draws, the speculative-demand-minus-supply interval changes from $[-55.3, 1.8]$ to $[-49.1, -9.3]$. Thus, shape information establishes that supply contributed more than speculation in posterior expectation on this reference reduced-form distribution. The tail-only interval on its own feasible draws is already negative, $[-53.6, -9.6]$; the common-draw comparison identifies the contribution of shape information, rather than a reversal of that model's historical interpretation. At $\kappa = (0.3, 0.3)$, neither contrast is signed even with degree-6 monotonicity.

Taken together, the three exhibits show that different economic claims require different identifying information. The collective importance of demand is relatively robust when the elasticity restrictions retain substantial credibility. The magnitude of individual variance and historical contributions is less tightly determined. Qualitative distributional information sharpens those magnitudes and, in selected comparisons, establishes a sign or ranking that the larger prior class leaves unresolved. The contribution of the framework is to make these distinctions explicit: it identifies which conclusions follow from support and probability restrictions alone, and which require additional beliefs about how probability is allocated across observationally equivalent structural models.

5 Conclusion

This paper develops a robust Bayesian framework for structural models in which economic knowledge restricts selected features of a prior distribution without determining the complete prior. This distinction is particularly important in set-identified models, where the likelihood cannot distinguish observationally equivalent structural parameters and the conditional structural prior is therefore not updated by the data. Rather than selecting

an arbitrary complete prior or allowing all distributions on the identified set, we impose economically interpretable restrictions on the marginal distributions of selected structural quantities, while leaving their remaining features and joint completion unrestricted. These restrictions may encode probability bounds, moments, or qualitative distributional shape information.

The framework exploits the likelihood structure of set-identified models to formulate posterior sensitivity analysis as linear optimization over conditional structural priors. We show that nonparametric shape restrictions, including decreasing marginal densities, admit affine representations and can therefore be incorporated into the same optimization framework. We establish conditions under which the resulting infinite-dimensional bounds can be consistently approximated by finite-dimensional linear programs that reweight simulated structural draws. The conditional construction induces a corresponding class of joint priors, while a complementary specialization permits economically meaningful features of a baseline prior to be preserved through its conditional distribution given the restricted marginal.

The oil-market application illustrates why distinguishing different forms of prior information matters for substantive conclusions. Under conventional elasticity restrictions, the combined contribution of flow-demand and speculative-demand shocks exceeds that of supply shocks with high posterior probability uniformly over the admissible prior class. This conclusion survives moderate relaxation of the elasticity restrictions, whereas the stronger ranking that supply contributes less than either demand shock individually remains largely unresolved. At the same time, the conventional restrictions leave considerable ambiguity about the quantitative contributions of individual shocks. Imposing decreasing marginal densities on the elasticities substantially narrows posterior expectation bounds for forecast-error variance shares and historical decompositions, even when all prior mass already satisfies the conventional elasticity bounds. Moreover, these gains persist when an additional hard upper bound of 0.15 is imposed on supply elasticities. Thus, restricting the support of elasticities does not exhaust the identifying information contained in their prior distributions. These com-

parisons hold the feasible reduced-form draws fixed, isolating the incremental information supplied by the distributional restrictions.

The broader implication is that different economic conclusions require different forms and strengths of prior information. By treating the economically defensible features of a prior as the primitive identifying assumptions, the proposed framework distinguishes conclusions supported by those features from conclusions that depend on how otherwise unspecified prior beliefs are completed. This provides a systematic way to incorporate partial economic knowledge while remaining robust to the aspects of the prior that the researcher cannot credibly specify.

References

Alonso-Alvarez, I., V. Di Nino, and F. Venditti (2022): "Strategic Interactions and Price Dynamics in the Global Oil Market," *Energy Economics*, 107, 105739.

Antolin-Diaz, J. and J. Rubio-Ramirez (2018): "Narrative Sign Restrictions for SVARs," *The American Economic Review*, 108, 2802–2829.

Arias, J. Rubio-Ramirez, J. and D. Waggoner (2018): "Inference based on Structural Vector Autoregressions identified with Sign and Zero Restrictions: Theory and Applications," *Econometrica*, 86, 685–720.

——— (2024): "Uniform Priors for Impulse Responses," *forthcoming*.

Baumeister, C. and J. Hamilton (2015): "Sign Restrictions, Structural Vector Autoregressions, and Useful Prior Information," *Econometrica*, 83, 1963–1999.

——— (2019): "Structural Interpretation of Vector Autoregressions with Incomplete Identification: Revisiting the Role of Oil Supply and Demand Shocks," *American Economic Review*, 109, 1873–1910.

——— (2020): "Structural Interpretation of Vector Autoregressions with Incomplete Identification: Setting the Record Straight," Working paper, University of California at San Diego.

——— (2023): "Uncovering Disaggregated Oil Market Dynamics: A Full-Information Approach to Granular Instrumental Variables," Working paper, revised June 2024.

Caldara, D., M. Cavallo, and M. Iacoviello (2019): "Oil Price Elasticities and Oil Price Fluctuations," *Journal of Monetary Economics*, 103, 1–20.

Canay, I. A. and A. M. Shaikh (2017): "Practical and Theoretical Advances in Inference for Partially Identified Models," in *Advances in Economics and Econometrics: Eleventh World Congress*, Cambridge University Press, vol. 2, 271–306.

Gazzani, A., F. Venditti, and G. Veronese (2024): "Oil Price Shocks in Real Time," *Journal of Monetary Economics*, 144, 103547.

Giacomini, R. and T. Kitagawa (2021): "Robust Bayesian Inference for Set-Identified Models," *Econometrica*, 89, 1519–1556.

Giacomini, R., T. Kitagawa, and H. Uhlig (2019): "Estimation under Ambiguity," Working Paper CWP24/19, Centre for Microdata Methods and Practice (cemmap).

Giacomini, R., T. Kitagawa, and A. Volpicella (2022): "Uncertain Identification," *Quantitative Economics*, 12, 95–123.

Gustafson, P. (2014): "Bayesian inference in partially identified models: Is the shape of the posterior distribution useful?" *Electronic Journal of Statistics*, 8, 476–496.

Herrera, A. and S. Rangaraju (2020): "The Effect of Oil Supply Shocks on U.S. Economic Activity: What Have We Learned?" *Journal of Applied Econometrics*, 35, 141–159.

Ho, P. (2023): "Global Robust Bayesian Analysis in Large Models," *Journal of Econometrics*, 235, 608–642.

Känzig, D. R. (2021): "The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements," *American Economic Review*, 111, 1092–1125.

Kessler, D., P. Hoff, and D. Dunson (2015): "Marginally specified priors for nonparametric Bayesian estimation," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 77, 35–58.

Kilian, L. (2009): "Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market," *American Economic Review*, 99, 1053–69.

——— (2022): "Facts and Fiction in Oil Market Modeling," *Energy Economics*, 110, 105973.

Kilian, L. and D. Murphy (2012): "Why Agnostic Sign Restrictions Are Not Enough: Understanding the Dynamics of Oil Market VAR Models," *Journal of European Economic Association*, 10, 1166–88.

——— (2014): "The Role of Inventories and Speculative Trading in the Global Market for Crude Oil," *Journal of Applied Econometrics*, 29, 454–78.

Kilian, L. and X. Zhou (2019): "Oil Supply Shock Redux?" Working paper, University of Michigan.

Lam, H., Z. Liu, and D. I. Singham (2024): "Shape-Constrained Distributional Optimization via Importance-Weighted Sample Average Approximation," *arXiv preprint arXiv:2406.07825*.

Lavine, M., L. Wasserman, and R. Wolpert (1991): "Bayesian inference with specified marginals," *Journal of the American Statistical Association*, 86, 964–971.

Liu, Y., A. Pichler, and H. Xu (2019): "Discrete Approximation and Quantification in Distributionally Robust Optimization," *Mathematics of Operations Research*, 44, 19–37.

Manski, C. (2003): *Partial Identification of Probability Distributions*, Springer.

Masten, M. A. and A. Poirier (2020): "Inference on Breakdown Frontiers," *Quantitative Economics*, 11, 41–111.

Molinari, F. (2020): "Microeconometrics with Partial Identification," in *Handbook of Econometrics*, Elsevier, vol. 7A, 355–486.

Moon, H. and F. Schorfheide (2012): "Bayesian and Frequentist Inference in Partially Identified Models," *Econometrica*, 80, 755–782.

Popescu, I. (2005): "A Semidefinite Programming Approach to Optimal-Moment Bounds for Convex Classes of Distributions," *Mathematics of Operations Research*, 30, 632–657.

Read, M. and D. Zhu (2025): "Fast Posterior Sampling in Tightly Identified SVARs Using 'Soft' Sign Restrictions," Research Discussion Paper 2025-03, Reserve Bank of Australia.

Rubio-Ramirez, J., D. Waggoner, and T. Zha (2010): "Structural Vector Autoregressions: Theory of Identification and Algorithms for Inference," *The Review of Economic Studies*, 77.

Uhlig, H. (2005): "What are the effects of monetary policy on output? Results from an agnostic identification procedure," *Journal of Monetary Economics*, 52, 381–419.

Zhou, X. (2020): "Refining the Workhorse Oil Market Model," *Journal of Applied Econometrics*, 35, 130–140.

A Proofs

A.1 Proof of Proposition 1

Proof. Fix ϕ and let $\pi \in \Pi_{\mathcal{M},\mathcal{R}}(\phi)$. Under Assumption 1, the likelihood is constant in θ conditional on ϕ. Hence, for every ϕ for which the conditional posterior is well defined,

$$\begin{aligned}\pi(d\theta \mid Y, \phi) &= \frac{p(Y \mid \phi, \theta)\, \pi(d\theta)}{\int_\Theta p(Y \mid \phi, \tilde{\theta})\, \pi(d\tilde{\theta})} \\ &= \frac{p(Y \mid \phi)\, \pi(d\theta)}{p(Y \mid \phi) \int_\Theta \pi(d\tilde{\theta})} = \pi(d\theta).\end{aligned}$$

Therefore,

$$\mathbb{E}_\pi[T \mid Y, \phi] = \int_\Theta T(\theta)\, \pi(d\theta).$$

Since T is bounded and measurable, this integral is finite for every $\pi \in \Pi_{\mathcal{M},\mathcal{R}}(\phi)$. Taking the infimum and supremum over the admissible prior class gives

$$\begin{aligned}\underline{m}_T(\phi) &= \inf_{\pi \in \Pi_{\mathcal{M},\mathcal{R}}(\phi)} \int_\Theta T(\theta)\, \pi(d\theta), \\ \overline{m}_T(\phi) &= \sup_{\pi \in \Pi_{\mathcal{M},\mathcal{R}}(\phi)} \int_\Theta T(\theta)\, \pi(d\theta).\end{aligned}$$

Now suppose that the admissible prior class has the affine representation

$$\Pi_{\mathcal{M},\mathcal{R}}(\phi) = \left\{ \pi \in \mathcal{P}(\mathrm{IS}_\theta(\phi)) : \int a_j(\theta)\, \pi(d\theta) \le b_j \quad \forall j \in J \right\}.$$

The map

$$\pi \mapsto \int_\Theta T(\theta)\, \pi(d\theta)$$

is linear in the decision variable π, and each constraint

$$\int_\Theta a_j(\theta)\,\pi(d\theta) \le b_j$$

is affine in π. The probability-measure restrictions $\pi \ge 0$ and $\int_\Theta \pi(d\theta) = 1$ are likewise linear. Consequently, the lower and upper bounds are the optimal values of, respectively, an infinite-dimensional linear minimization and maximization problem over probability measures. □

A.2 Auxiliary lemmas for Theorem 1

Fix ϕ throughout, write $K = \mathrm{IS}_\theta(\phi)$, and suppress dependence on ϕ. Let $\mathcal{C}$ be the finite partition generated by the restriction events G_ℓ and, for a target $T = \mathbf{1}_A$, the target event A. For a bounded function $f : K \to \mathbb{R}$, write

$$\omega_f(t) = \sup\big\{|f(x) - f(y)| : x, y \in K,\ d(x,y) \le t\big\}.$$

For the first N draws, define

$$h_N = \max_{C\in\mathcal{C}} \sup_{x\in C} \min_{\substack{1\le n\le N \\ \theta^{(n)}\in C}} d(x, \theta^{(n)}), \qquad \min \varnothing = +\infty.$$

The draws are defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Because K is a compact metric space, $\mathcal{P}(K)$ is compact and metrizable under weak convergence. We use the Portmanteau theorem: if $\pi_k \Rightarrow \pi$, then $\limsup_k \pi_k(F) \le \pi(F)$ for relatively closed $F \subseteq K$, and $\liminf_k \pi_k(G) \ge \pi(G)$ for relatively open $G \subseteq K$. Consequently, $\pi \mapsto \pi(F)$ is weakly upper semicontinuous, $\pi \mapsto \pi(G)$ is weakly lower semicontinuous, and $\pi \mapsto \int_K f\,d\pi$ is weakly continuous for $f \in C(K)$.

Lemma 1 (Closedness and nonemptiness)**.** *Under Assumptions 2 and 3, each $\Pi^{(r)}(\phi)$ and $\Pi(\phi)$ is weakly compact. If also Assumptions 4 and 6 hold, then $\Pi(\phi) \neq \varnothing$.*

Proof. For every $j \in J$, the set $\{\pi : \int_K a_j \, d\pi \leq b_j\}$ is weakly closed because $a_j \in C(K)$, so that $\pi \mapsto \int_K a_j \, d\pi$ is weakly continuous by the definition of weak convergence. For every $\ell \in L$, the set $\{\pi : \pi(G_\ell) \leq \kappa_\ell\}$ is weakly closed because G_ℓ is relatively open, so that $\pi \mapsto \pi(G_\ell)$ is weakly lower semicontinuous by the Portmanteau theorem; since $\mathcal{P}(K)$ is metrizable, sequential lower semicontinuity suffices. Intersections of these closed sets are closed. Since K is a compact metric space, $\mathcal{P}(K)$ is weakly compact by Prokhorov's theorem, so both the population and stage classes, as closed subsets of $\mathcal{P}(K)$, are compact.

Assumption 6 makes every stage class nonempty. By Assumption 4, these compact classes are decreasing and their intersection is $\Pi(\phi)$; the finite intersection property therefore gives $\Pi(\phi) \neq \varnothing$. □

Lemma 2 (Measurability of the finite-program values)**.** *Under Assumptions 3, 4, and 5, for $T \in C(K)$ or $T = \mathbf{1}_A$ with A relatively closed, the bounds $\bar{m}_{T,N,r}(\phi)$ and $\underline{m}_{T,N,r}(\phi)$ are extended-real random variables for every N, r.*

Proof. Fix N, r, and let β_r collect the right-hand sides $(b_j)_{j \in J_r}$ and $(\kappa_\ell)_{\ell \in L}$. For $c \in \mathbb{R}^N$ and a matrix M with N columns and $|J_r| + |L|$ rows, set

$$V(c, M) = \max\{c^\top w : w \in \Delta_N, \ Mw \leq \beta_r\}, \qquad V(c, M) = -\infty \text{ if infeasible.}$$

The map V is upper semicontinuous. Indeed, if $(c_k, M_k) \to (c, M)$ and $V(c_k, M_k) \geq \alpha$, choose a maximizer w_k for each k. A subsequence converges in the compact simplex Δ_N to some w. Passing to the limit gives $Mw \leq \beta_r$ and $c^\top w = \lim_k c_k^\top w_k \geq \alpha$. Thus every superlevel set $\{V \geq \alpha\}$ is closed, and V is Borel measurable.

The entries of the objective vector and restriction matrix are, respectively, $T(\theta^{(n)})$, $a_j(\theta^{(n)})$, and $\mathbf{1}_{G_\ell}(\theta^{(n)})$. They are measurable because the draws are random elements of K, the a_j are continuous, and both target types and the event indicators are Borel measurable. Therefore $\bar{m}_{T,N,r}(\phi) = V(c, M)$ is measurable, as is $\underline{m}_{T,N,r}(\phi) = -V(-c, M)$. □

Lemma 3 (Cell covering radius)**.** *Under Assumptions 2 and 5, there is* $\Omega_0 \in \mathcal{F}$ *with* $\mathbb{P}(\Omega_0) = 1$ *such that* $h_N \to 0$ *on* Ω_0*.*

Proof. Fix $k \in \mathbb{N}$ and $C \in \mathcal{C}$. Cover the compact set K by finitely many balls $B(y_i, 1/(4k))$. For every ball meeting C, choose $x_i \in B(y_i, 1/(4k)) \cap C$. The corresponding balls $B(x_i, 1/(2k))$ cover C. By Assumption 5, each set $B(x_i, 1/(2k)) \cap C$ has positive probability p_i. Independence gives probability $(1 - p_i)^N \to 0$ that none of the first N draws lies in this set. Hence, almost surely, each of the finitely many sets contains a draw eventually.

Once they have all been hit, every $x \in C$ is within $1/k$ of a draw in C. The cells are finite in number, so almost surely $h_N \leq 1/k$ eventually. Intersect these measurable probability-one events over $k \in \mathbb{N}$ to obtain Ω_0; on this event $h_N \to 0$. □

Lemma 4 (Cell-preserving projection)**.** *Fix a realization of the draws and* N *with* $h_N < \infty$*. For* x *in cell* C*, let* $\sigma_N(x)$ *be a closest draw among* $\{\theta^{(n)} : n \leq N,\ \theta^{(n)} \in C\}$*, breaking ties by the smallest index. Then* $\sigma_N : K \to \Theta_N$ *is Borel measurable. For any* $\pi \in \mathcal{P}(K)$*, its pushforward* $P_N = \pi \circ \sigma_N^{-1} \in \mathcal{P}(\Theta_N)$ *satisfies:*

(P1) $\sigma_N(x)$ *lies in the same cell as* x*;*

(P2) $d(x, \sigma_N(x)) \leq h_N$ *for every* $x \in K$*;*

(P3) $P_N(E) = \pi(E)$ *for every union* E *of cells, including each* G_ℓ *and, for an indicator target,* A*;*

(P4) For every bounded Borel function $f : K \to \mathbb{R}$*,*

$$\left| \int_K f \, dP_N - \int_K f \, d\pi \right| \leq \omega_f(h_N).$$

Proof. For each cell C, put $I_C = \{n \leq N : \theta^{(n)} \in C\}$, which is nonempty because $h_N < \infty$.

For $n \in I_C$, define the set of points assigned *index* n by

$$H_{n,C} = \big\{ x \in C : d(x, \theta^{(n)}) < d(x, \theta^{(m)}) \quad \text{for all } m \in I_C,\ m < n; \\ d(x, \theta^{(n)}) \le d(x, \theta^{(m)}) \quad \text{for all } m \in I_C,\ m > n \big\}.$$

These sets are Borel and partition C; on $H_{n,C}$, $\sigma_N(x) = \theta^{(n)}$. Thus σ_N is Borel even when two draws have the same location. Properties (P1) and (P2) follow from its construction. For any union E of cells, (P1) implies $\sigma_N^{-1}(E) = E$, proving (P3). The restriction events and, when applicable, A are unions of cells by their definition. Finally, the pushforward identity and (P2) give

$$\left| \int_K f \, dP_N - \int_K f \, d\pi \right| = \left| \int_K (f \circ \sigma_N - f) \, d\pi \right| \le \omega_f(h_N),$$

proving (P4). □

Lemma 5 (Deterministic diagonal)**.** *Let $X_{N,r}$ be extended-real random variables and $x_r, x \in \mathbb{R}$. Suppose $X_{N,r} \to x_r$ almost surely as $N \to \infty$ for every fixed r, and $x_r \to x$. Then some deterministic nondecreasing $r(N) \to \infty$ satisfies $X_{N,r(N)} \to x$ almost surely. One schedule can serve finitely many such arrays simultaneously.*

Proof. Define $D_{M,r} = \sup_{N \ge M} |X_{N,r} - x_r|$, allowing $D_{M,r} = +\infty$. This is measurable, and for each fixed r, $D_{M,r} \to 0$ almost surely, hence in probability, as $M \to \infty$. Choose deterministic $N_1 < N_2 < \cdots$ such that $\mathbb{P}(D_{N_r,r} > 1/r) \le 2^{-r}$, and set

$$r(N) = \begin{cases} 1, & N < N_1, \\ \max\{r : N_r \le N\}, & N \ge N_1. \end{cases}$$

By the Borel–Cantelli lemma, almost surely $D_{N_r,r} \le 1/r$ for all sufficiently large r. Hence,

on that event and for all sufficiently large N,

$$|X_{N,r(N)} - x| \le D_{N_{r(N)},r(N)} + |x_{r(N)} - x| \le \frac{1}{r(N)} + |x_{r(N)} - x| \longrightarrow 0.$$

For finitely many arrays, choose each N_r so that the sum of their exceptional probabilities is at most 2^{-r}. □

A.3 Proof of Theorem 1

Proof. Fix ϕ, write $K = \mathrm{IS}_\theta(\phi)$, and suppress dependence on ϕ where no confusion arises. Let

$$\bar{m}^{(r)} = \sup_{\pi \in \Pi^{(r)}} \int_K T \, d\pi, \qquad \underline{m}^{(r)} = \inf_{\pi \in \Pi^{(r)}} \int_K T \, d\pi.$$

Write $\bar{m}_{N,r}$ and $\underline{m}_{N,r}$ for the sampled bounds, and $\bar{m}$ and $\underline{m}$ for the population bounds. Whenever $h_N < \infty$, set

$$e_{N,r} = \max_{j \in J_r} \omega_{a_j}(h_N), \qquad \max \varnothing = 0,$$

and

$$\Delta_{N,r} = \begin{cases} \omega_T(h_N) + 2\|T\|_\infty \dfrac{e_{N,r}}{\eta_r}, & T \in C(K), \\ \dfrac{e_{N,r}}{\eta_r}, & T = \mathbf{1}_A. \end{cases}$$

We take $e_{N,r}/\eta_r = 0$ when $J_r = \varnothing$.

The argument proceeds in five steps. We first compare the sampled and population feasible sets at a fixed stage and bound the resulting error. We then pass to the limits in N and r, in that order, before using a diagonal argument to obtain the joint limit.

Step 1: Every $\pi \in \Pi_{N,r}$ is a probability measure supported on Θ_N that satisfies all stage-r restrictions. Hence $\Pi_{N,r} \subseteq \Pi^{(r)}$, and therefore

$$\bar{m}_{N,r} \leq \bar{m}^{(r)}, \qquad \underline{m}_{N,r} \geq \underline{m}^{(r)}.$$

Step 2: Fix r, N, and a realization of the draws such that $h_N < \infty$ and $e := e_{N,r} \leq \eta_r$. Let $\pi \in \Pi^{(r)}$ be arbitrary, and let $\pi_r^\circ \in \Pi^{(r)}$ be the prior supplied by Assumption 6. Apply the cell-preserving projection of Lemma 4 to obtain

$$P_N = \pi \circ \sigma_N^{-1}, \qquad P_N^\circ = \pi_r^\circ \circ \sigma_N^{-1}.$$

If $J_r \neq \varnothing$, put $t = e/\eta_r \in [0,1]$. If $J_r = \varnothing$, put $t = 0$. In either case, define

$$Q_N = (1-t)P_N + tP_N^\circ \in \mathcal{P}(\Theta_N).$$

Suppose first that $J_r \neq \varnothing$. For each $j \in J_r$, property (P4) of Lemma 4 and the Slater margin give

$$\int_K a_j \, dP_N \leq b_j + e, \qquad \int_K a_j \, dP_N^\circ \leq b_j - \eta_r + e.$$

Consequently,

$$\int_K a_j \, dQ_N \leq (1-t)(b_j + e) + t(b_j - \eta_r + e) = b_j + e - t\eta_r = b_j.$$

When $J_r = \varnothing$, there are no continuous rows to check.

For every event row, property (P3) gives

$$Q_N(G_\ell) = (1-t)\pi(G_\ell) + t\pi_r^\circ(G_\ell) \leq \kappa_\ell,$$

because π and π_r° both belong to $\Pi^{(r)}$. Thus $Q_N \in \Pi_{N,r}$; in particular, the sampled stage

class is nonempty.

For the objective,

$$\int_K T\,dQ_N - \int_K T\,d\pi = (1-t)\left(\int_K T\,dP_N - \int_K T\,d\pi\right) + t\left(\int_K T\,dP_N^\circ - \int_K T\,d\pi\right).$$

If T is continuous, (P4) bounds the absolute value of the first bracket by $\omega_T(h_N)$, while the second is bounded by $2\|T\|_\infty$. If $T = \mathbf{1}_A$, (P3) makes the first bracket zero, and the second equals $\pi_r^\circ(A) - \pi(A) \in [-1, 1]$. Hence, in both cases,

$$\left|\int_K T\,dQ_N - \int_K T\,d\pi\right| \le \Delta_{N,r}.$$

Since Q_N is sampled-stage feasible, this inequality holds for a possibly different Q_N for every $\pi \in \Pi^{(r)}$. Taking the supremum and infimum over π, respectively, yields

$$\bar{m}_{N,r} \ge \bar{m}^{(r)} - \Delta_{N,r}, \qquad \underline{m}_{N,r} \le \underline{m}^{(r)} + \Delta_{N,r}.$$

Combining these inequalities with Step 1 gives

$$0 \le \bar{m}^{(r)} - \bar{m}_{N,r} \le \Delta_{N,r}, \qquad 0 \le \underline{m}_{N,r} - \underline{m}^{(r)} \le \Delta_{N,r}.$$

Step 3: Let Ω_0 be the probability-one event in Lemma 3. On Ω_0, $h_N \to 0$, so $h_N < \infty$ for all sufficiently large N. For each fixed r, the set J_r is finite and every a_j is uniformly continuous on the compact space K. Therefore $e_{N,r} \to 0$ on Ω_0. Since $\eta_r > 0$ when $J_r \neq \varnothing$, the condition $e_{N,r} \le \eta_r$ holds eventually.

For a continuous target, uniform continuity also gives $\omega_T(h_N) \to 0$. For an indicator target, $\Delta_{N,r} = e_{N,r}/\eta_r \to 0$ directly; projection preserves the probability of A exactly. The

error bound in Step 2 therefore implies, for every fixed r,

$$\bar{m}_{N,r} \longrightarrow \bar{m}^{(r)}, \qquad \underline{m}_{N,r} \longrightarrow \underline{m}^{(r)} \quad \text{almost surely as } N \to \infty.$$

The same event Ω_0 works for all fixed stages because the cell partition does not depend on r.

Step 4: Lemma 1 gives $\Pi \neq \varnothing$ and compactness of every stage class. By Assumption 4,

$$\Pi \subseteq \Pi^{(r+1)} \subseteq \Pi^{(r)}.$$

Thus $\bar{m}^{(r)}$ decreases toward a limit no smaller than $\bar{m}$, while $\underline{m}^{(r)}$ increases toward a limit no greater than $\underline{m}$.

Consider first the upper bound. The map $\Phi(\pi) = \int_K T\, d\pi$ is weakly continuous when $T \in C(K)$. When $T = \mathbf{1}_A$ with A relatively closed, it is weakly upper semicontinuous by Portmanteau. Hence it attains its maximum on each compact $\Pi^{(r)}$; choose $\pi_r \in \Pi^{(r)}$ with $\Phi(\pi_r) = \bar{m}^{(r)}$. Compactness of $\mathcal{P}(K)$ gives a weakly convergent subsequence $\pi_{r_k} \Rightarrow \pi^*$. For every fixed s, the nesting of the stage classes implies $\pi_{r_k} \in \Pi^{(s)}$ for all sufficiently large k. Because $\Pi^{(s)}$ is weakly closed,

$$\pi^* \in \bigcap_{s \geq 1} \Pi^{(s)} = \Pi.$$

Upper semicontinuity now gives

$$\lim_{r \to \infty} \bar{m}^{(r)} = \lim_{k \to \infty} \Phi(\pi_{r_k}) \leq \Phi(\pi^*) \leq \bar{m}.$$

Together with $\bar{m}^{(r)} \geq \bar{m}$, this proves $\bar{m}^{(r)} \downarrow \bar{m}$. If T is continuous, applying the same argument to $-T$ also proves $\underline{m}^{(r)} \uparrow \underline{m}$.

It remains to consider the lower bound when $T = \mathbf{1}_A$. Let $\underline{m}_\infty = \lim_r \underline{m}^{(r)} \leq \underline{m}$. Choose

$\pi_r \in \Pi^{(r)}$ such that

$$\pi_r(A) \leq \underline{m}^{(r)} + \frac{1}{r}.$$

A subsequence converges weakly to some $\pi^* \in \Pi$ by the same compactness and nesting argument. Since $\mathrm{int}_K A$ is relatively open, Portmanteau gives

$$\pi^*(\mathrm{int}_K A) \leq \liminf_{k\to\infty} \pi_{r_k}(\mathrm{int}_K A) \leq \liminf_{k\to\infty} \pi_{r_k}(A) \leq \underline{m}_\infty.$$

Consequently,

$$\inf_{\pi\in\Pi} \pi(\mathrm{int}_K A) \leq \underline{m}_\infty \leq \underline{m}.$$

Assumption 7 makes the outer quantities equal, so $\underline{m}_\infty = \underline{m}$. The same conclusion holds without that assumption if $\Pi^{(r)} = \Pi$ for all sufficiently large r.

Step 5: By Lemma 2, the sampled bounds are extended-real random variables. Step 3 gives their almost sure limits for every fixed r, and Step 4 shows that those stage limits converge to the population bounds under the conditions of the theorem. Applying Lemma 5 simultaneously to the upper- and lower-bound arrays yields a deterministic nondecreasing sequence $r(N) \to \infty$ such that

$$\bar{m}_{N,r(N)} \longrightarrow \bar{m}, \qquad \underline{m}_{N,r(N)} \longrightarrow \underline{m} \quad \text{almost surely.}$$

□

A.4 Proof of Corollary 1

Proof. Fix ϕ. For any $\pi \in \mathcal{P}(K)$, let $p_C = \pi(C)$ for each $C \in \mathcal{C}$. Because the cells partition K, the vector $p = (p_C)_{C\in\mathcal{C}}$ belongs to $\Delta(\mathcal{C})$. Each restriction event G_ℓ and the target event

A is a union of cells, so

$$\pi(G_\ell) = \sum_{C \subseteq G_\ell} p_C, \qquad \pi(A) = \sum_{C \subseteq A} p_C.$$

Since $J = \varnothing$, these are all the restrictions on π. Thus every population-feasible prior produces a vector $p \in \mathcal{D}$ with the objective value stated in the corollary.

Conversely, for any $p \in \mathcal{D}$, choose one point $x_C \in C$ from each nonempty cell and define

$$\pi_p = \sum_{C \in \mathcal{C}} p_C \delta_{x_C}.$$

This prior belongs to $\Pi(\phi)$ and assigns mass p_C to each cell. Hence every vector in $\mathcal{D}$ is attained by a population-feasible prior. Taking the minimum and maximum over $\mathcal{D}$ proves the cell-mass formulas.

Now suppose that every cell contains at least one of the first N simulated points. Choose such a point $\theta_C \in C$ for each cell. For every $p \in \mathcal{D}$, the sampled prior

$$\pi_{N,p} = \sum_{C \in \mathcal{C}} p_C \delta_{\theta_C}$$

has the same cell masses, satisfies every event restriction, and has objective value $\sum_{C \subseteq A} p_C$. Conversely, the cell masses of every sampled feasible prior belong to $\mathcal{D}$. The sampled and population programs therefore optimize over the same vectors p, so both their lower and upper values are equal.

Finally, for each cell C,

$$\mathbb{P}\big(\theta^{(1)}, \ldots, \theta^{(N)} \notin C\big) = (1 - \nu(C))^N \longrightarrow 0.$$

There are finitely many cells and $\nu(C) > 0$ for each one. Almost surely, all cells have therefore been hit by some finite random time N_0. They remain represented for every $N \geq N_0$, giving

the asserted eventual equality. Since $J = \varnothing$, the stage index r has no effect. □

A.5 Proof of Proposition 2

Proof. For the first result, since $\mu_\phi = S_\# \pi_{\theta|\phi}$,

$$\begin{aligned}\pi(S(\theta) \in B) &= \int_\Phi \mu_\phi(B)\, \pi_\phi(d\phi) \\ &\leq \int_\Phi \kappa\, \pi_\phi(d\phi) = \kappa,\end{aligned}$$

where the inequality follows from $\mu_\phi(B) \leq \kappa$ for π_ϕ-almost every ϕ.

For the second result, let $\mu_{Z|\phi}$ denote the conditional distribution of Z given ϕ, extend $f_{Z|\phi}$ by zero outside $[0, \bar{z}(\phi)]$, and note that extension by zero preserves monotonicity on $(0, \infty)$. Since $f_{Z|\phi}$ is non-increasing on $(0, \infty)$, its averages over $(z, z + 1/m]$ converge to its right limit, so its right-continuous version is

$$f_{Z|\phi}(z) = \lim_{m \to \infty} m\, \mu_{Z|\phi}\big((z, z + 1/m]\big), \qquad z > 0.$$

For fixed z and m, the map $\phi \mapsto \mu_{Z|\phi}((z, z+1/m])$ is measurable because $\pi_{\theta|\phi}$ is a probability kernel, and for fixed ϕ it is right-continuous in z. It is therefore jointly measurable in (ϕ, z), and so is its limit. The weight $w(\phi) = \mu_{Z|\phi}(\{0\})$ is measurable for the same reason. The unconditional distribution of Z is the mixture of its conditional distributions over ϕ. Hence, by Tonelli's theorem, for every Borel set $B \subseteq [0, \infty)$,

$$\pi(Z \in B) = \bar{w}\, \mathbf{1}_B(0) + \int_B f_Z(z)\, dz, \qquad f_Z(z) = \int_\Phi f_{Z|\phi}(z)\, \pi_\phi(d\phi).$$

Moreover, for any $0 < z_1 \leq z_2$, $f_{Z|\phi}(z_1) \geq f_{Z|\phi}(z_2)$ for π_ϕ-almost every ϕ. Integrating this

inequality over ϕ gives

$$\begin{aligned} f_Z(z_1) &= \int_\Phi f_{Z|\phi}(z_1)\,\pi_\phi(d\phi) \\ &\geq \int_\Phi f_{Z|\phi}(z_2)\,\pi_\phi(d\phi) = f_Z(z_2), \end{aligned}$$

so f_Z is non-increasing on $(0,\infty)$. It is finite there, because a non-increasing density satisfies $f_{Z|\phi}(z) \leq 1/z$.

Finally, under Assumption 1 the conditional structural distribution given ϕ is not updated by the data, so $\pi_{\theta|Y,\phi} = \pi_{\theta|\phi}$ and therefore the conditional marginal μ_ϕ of S, and the corresponding conditional distribution $\mu_{Z|\phi}$ of Z, are unchanged. The joint posterior only replaces the mixing distribution π_ϕ by $\pi_\phi(\cdot \mid Y)$. Repeating the preceding arguments with $\pi_\phi(d\phi \mid Y)$ therefore proves both posterior statements. □

A.6 Proof of Proposition 3

Proof. Fix $\mu \in \mathcal{M}_S(\phi)$ and let $r_{\mu,\phi} = d\mu/d\pi^0_{S|\phi}$. Since $\mathcal{R}(\phi) = \{\rho^0_\phi\}$, the unique prior with marginal μ and baseline completion is

$$\pi^\mu_{\theta|\phi}(A) = \int \rho^0_\phi(A \mid s)\,\mu(ds), \qquad A \in \mathcal{A}_\theta.$$

Uniqueness follows directly because both the marginal μ and the completion ρ^0_ϕ are fixed.

Using $\mu(ds) = r_{\mu,\phi}(s)\pi^0_{S|\phi}(ds)$ and the disintegration of the baseline prior,

$$\begin{aligned} \pi^\mu_{\theta|\phi}(A) &= \int \rho^0_\phi(A \mid s) r_{\mu,\phi}(s)\,\pi^0_{S|\phi}(ds) \\ &= \int \mathbf{1}_A(\theta) r_{\mu,\phi}(S(\theta))\,\pi^0_{\theta|\phi}(d\theta). \end{aligned}$$

Hence

$$\pi^{\mu}_{\theta|\phi}(d\theta) = \frac{d\mu}{d\pi^{0}_{S|\phi}}(S(\theta))\,\pi^{0}_{\theta|\phi}(d\theta).$$

In particular, $\pi^{\mu}_{\theta|\phi} \ll \pi^{0}_{\theta|\phi}$, and therefore

$$\begin{aligned}
\mathrm{KL}\left(\pi^{\mu}_{\theta|\phi} \,\|\, \pi^{0}_{\theta|\phi}\right) &= \int \log r_{\mu,\phi}(S(\theta))\,\pi^{\mu}_{\theta|\phi}(d\theta) \\
&= \int \log r_{\mu,\phi}(s)\,\mu(ds) \\
&= \mathrm{KL}\left(\mu \,\|\, \pi^{0}_{S|\phi}\right).
\end{aligned}$$

Finally, let $q_{\theta|\phi} \ll \pi^{0}_{\theta|\phi}$ satisfy $S_{\#}q_{\theta|\phi} = \mu$. By the chain rule for relative entropy,

$$\begin{aligned}
\mathrm{KL}\left(q_{\theta|\phi} \,\|\, \pi^{0}_{\theta|\phi}\right) &= \mathrm{KL}\left(\mu \,\|\, \pi^{0}_{S|\phi}\right) \\
&\quad + \int \mathrm{KL}\left(q_{\theta|S,\phi}(\cdot \mid s) \,\|\, \rho^{0}_{\phi}(\cdot \mid s)\right)\mu(ds).
\end{aligned}$$

The second term is nonnegative and is zero for $\pi^{\mu}_{\theta|\phi}$, whose conditional distribution given $S = s$ is $\rho^{0}_{\phi}(\cdot \mid s)$. Therefore, whenever the marginal divergence is finite, $\pi^{\mu}_{\theta|\phi}$ minimizes KL divergence over all $q_{\theta|\phi} \ll \pi^{0}_{\theta|\phi}$ with marginal μ.

If another $q_{\theta|\phi}$ attains the same minimum, the conditional KL term must vanish for μ-almost every s, so

$$q_{\theta|S,\phi}(\cdot \mid s) = \rho^{0}_{\phi}(\cdot \mid s) \qquad \mu\text{-a.s.}$$

Together with $S_{\#}q_{\theta|\phi} = \mu$, this implies $q_{\theta|\phi} = \pi^{\mu}_{\theta|\phi}$, establishing uniqueness. □

A.7 Proof Toy Example

Proof. Under $\Pi^{\downarrow}_{\theta|\phi}$ we can write

$$\theta = \phi + \lambda U,$$

where $U \in [0,1]$ has an absolutely continuous distribution with density f_U that is weakly decreasing on $[0,1]$. Since the likelihood depends only on ϕ, the conditional posterior of θ given ϕ coincides with the conditional prior, so the set of posterior means is the set of prior means

$$\left\{\mathbb{E}[\theta|\phi] :\ \pi_{\theta|\phi} \in \Pi^{\downarrow}_{\theta|\phi}\right\} = \left\{\phi + \lambda\mathbb{E}[U] :\ f_U \text{ weakly decreasing on } [0,1]\right\}.$$

It therefore suffices to characterize the possible values of $\mathbb{E}[U]$. Let F_U denote the cdf of U. Since f_U is weakly decreasing,

$$F_U(u) = \int_0^u f_U(t)\,dt \geq u\int_0^1 f_U(t)\,dt = u \qquad \text{for all } u \in [0,1],$$

where the inequality follows because the average of a decreasing density over $[0,u]$ is at least its average over $[0,1]$. Hence $F_U(u) \geq u$, i.e. U is first-order stochastically dominated by the uniform distribution on $[0,1]$. Therefore

$$\mathbb{E}[U] = \int_0^1 (1 - F_U(u))\,du \leq \int_0^1 (1-u)\,du = \frac{1}{2}.$$

So every admissible posterior mean satisfies

$$\mathbb{E}[\theta|\phi] \leq \phi + \frac{\lambda}{2}.$$

This upper bound is attained by the uniform density $f_U(u) \equiv 1$, which belongs to $\Pi^{\downarrow}_{\theta|\phi}$.

For the lower bound, $\mathbb{E}[U] \geq 0$ always. Moreover, for $n \geq 1$, the decreasing densities

$$f_n(u) = n\,1\{0 \leq u \leq 1/n\}$$

belong to $\Pi^{\downarrow}_{\theta|\phi}$ and satisfy

$$\mathbb{E}_n[U] = \int_0^{1/n} un\,du = \frac{1}{2n} \to 0.$$

Hence $\inf \mathbb{E}[U] = 0$, so the closure of the set of posterior means is

$$\left[\phi, \phi + \frac{\lambda}{2}\right].$$

If one enlarges $\Pi^{\downarrow}_{\theta|\phi}$ to include weak limits of such priors, then the set itself equals $[\phi, \phi + \lambda/2]$. □

B Shape Restrictions

This appendix shows that the monotonicity restrictions of Example 7 and Section 4 can be represented by countably many affine restrictions with continuous integrands, via the Hausdorff moment problem. For integers $n \geq 0$ and $k \geq 1$, let $p_{n,k}(x) = x^{n+1}(1-x)^k$ on $[0,1]$, so that $q_{n,k} = p'_{n,k}$ in Example 7. Note that $p_{n,k} \geq 0$, $p_{n,k}(0) = p_{n,k}(1) = 0$, and $q_{n,k}(0) = \mathbf{1}\{n = 0\}$.

Lemma 6. *For $\mu \in \mathcal{P}([0,1])$ the following are equivalent:*

(a) $\mu(dx) = w_0\,\delta_0(dx) + f(x)\,dx$ for some $w_0 \in [0,1]$ and some weakly decreasing $f : (0,1] \to [0,\infty)$;

(b) $\int q_{n,k}\,d\mu \geq 0$ for all $n \geq 0$ and $k \geq 1$.

In particular, the set of distributions satisfying (a) is weakly closed. It is the weak closure of the set of distributions on $[0,1]$ with a weakly decreasing density.

Proof. (a) implies (b). Since f is weakly decreasing, for every $t \geq 0$ the set $\{x \in (0,1] : f(x) > t\}$ is an interval with left endpoint 0 and some right endpoint $a(t) \in [0,1]$. Because $\int_0^a q_{n,k}(x)\,dx = p_{n,k}(a)$, $q_{n,k}$ is bounded, and f is integrable, Fubini's theorem gives

$$\int_0^1 q_{n,k}(x) f(x)\,dx = \int_0^\infty \int_0^1 q_{n,k}(x)\,\mathbf{1}\{f(x) > t\}\,dx\,dt = \int_0^\infty p_{n,k}\big(a(t)\big)\,dt \geq 0.$$

Hence $\int q_{n,k}\,d\mu = w_0\,q_{n,k}(0) + \int_0^1 q_{n,k}(x) f(x)\,dx \geq 0$.

(b) implies (a). Let $m_n = (n+1)\int x^n\,\mu(dx)$. Expanding $p_{n,k}$ gives

$$q_{n,k}(x) = \sum_{i=0}^{k} (-1)^i \binom{k}{i} (n+1+i)\,x^{n+i}, \qquad \text{hence} \qquad \int q_{n,k}\,d\mu = \sum_{i=0}^{k} (-1)^i \binom{k}{i} m_{n+i}.$$

By (b), these iterated differences are nonnegative for all $n \geq 0$ and $k \geq 1$, and for $k = 0$ they equal $m_n \geq 0$. Thus $(m_n)_{n\geq 0}$ is completely monotone, and Hausdorff's moment theorem gives a probability measure λ on $[0,1]$ with $\int w^n\,\lambda(dw) = m_n$ for all n; its total mass is $m_0 = 1$. Let $W \sim \lambda$ and let U be uniform on $[0,1]$ and independent of W. Then $\mathbb{E}[(UW)^n] = m_n/(n+1) = \int x^n\,\mu(dx)$ for all n. Since distributions on $[0,1]$ are determined by their moments, μ is the distribution of UW. For a Borel set $B \subseteq (0,1]$, Tonelli's theorem therefore gives

$$\mu(B) = \mathbb{E}\left[\mathbf{1}\{W > 0\} \int_0^1 \mathbf{1}_B(uW)\,du\right] = \mathbb{E}\left[\mathbf{1}\{W > 0\}\,W^{-1} \int_0^W \mathbf{1}_B(x)\,dx\right] = \int_B f(x)\,dx,$$

with $f(x) = \mathbb{E}[W^{-1}\mathbf{1}\{W \geq x\}]$, which is weakly decreasing on $(0,1]$. Moreover, $\mu(\{0\}) = \mathbb{P}(W = 0) =: w_0$.

Closure. By (b), the set in question is an intersection of the sets $\{\mu : \int q_{n,k}\,d\mu \geq 0\}$, which are weakly closed because each $q_{n,k}$ is continuous. It contains every distribution with a weakly decreasing density. Conversely, if μ satisfies (a), then $\mu_j(dx) = \big(w_0 j\,\mathbf{1}\{0 < x \leq 1/j\} + f(x)\big)dx$ has a weakly decreasing density and $\mu_j \Rightarrow \mu$ as $j \to \infty$. □

Let $S : K_\phi \to [0, B]$ be continuous. Applying Lemma 6 to the distribution of S/B under

π shows that $S_{\#}\pi \in \mathcal{M}_S(\phi)$ in Example 7 if and only if

$$\int q_{n,k}\big(S(\theta)/B\big)\,\pi(d\theta) \geq 0 \qquad \text{for all } n \geq 0,\ k \geq 1.$$

These are countably many restrictions whose integrands are continuous on K_ϕ, as required by Assumption 3. Any increasing sequence of finite sets J_r with $\bigcup_r J_r = J$ then satisfies Assumption 4.

Monotonicity above a threshold is covered by the same lemma. Let $S : K_\phi \to (-\infty, B]$ be continuous, let $c < B$, and set $X = \max\{(S-c)/(B-c), 0\} \in [0,1]$. The distribution of X on $(0,1]$ is that of $(S-c)/(B-c)$ on $(0,1]$, while all prior mass on $\{S \leq c\}$ is mapped to the atom at zero, which Lemma 6 leaves unrestricted. Hence the distribution of S restricted to $(c, B]$ has a weakly decreasing density if and only if $\int q_{n,k}\big(X(\theta)\big)\,\pi(d\theta) \geq 0$ for all $n \geq 0$ and $k \geq 1$. Again, the integrands are continuous on K_ϕ.

Windowed rows for unbounded statistics. The rows above require a known bound B on S. The elasticities in Section 4 are ratios whose denominators can vanish on K_ϕ, so no such bound is available. Windowed rows do not require one. Fix windows $B_1 < B_2 < \cdots$ with $B_m \to \infty$ and, for $n \geq 0$ and $k \geq 2$, define $\tilde{q}_{n,k}$ on $[0,\infty]$ by $\tilde{q}_{n,k}(x) = q_{n,k}(x)$ for $x \leq 1$ and $\tilde{q}_{n,k}(x) = 0$ for $x > 1$. Since $q_{n,k}(1) = 0$ for $k \geq 2$, each $\tilde{q}_{n,k}$ is continuous on $[0,\infty]$.

Lemma 7. *Let Y be a random variable with values in $[-\infty,\infty]$ and law λ, let $c < B_1$, and set $X_m = (Y-c)^+/(B_m - c) \in [0,\infty]$. The following are equivalent:*

(a) the restriction of λ to (c,∞) is $f(y)\,dy$ for some weakly decreasing $f : (c,\infty) \to [0,\infty)$;

(b) $\mathbb{E}\,\tilde{q}_{n,k}(X_m) \geq 0$ for all $m \geq 1$, $n \geq 1$ and $k \geq 2$.

Under (a), the inequalities in (b) also hold for $n = 0$. Neither condition restricts λ on $[-\infty, c]$ or the mass $\lambda(\{+\infty\})$.

Proof. Fix m and let ν_m be the restriction of the law of X_m to $(0,1)$. Since $\tilde{q}_{n,k}$ vanishes on

$[1,\infty]$ for $k \geq 2$ and $q_{n,k}(0) = \mathbf{1}\{n=0\}$,

$$\mathbb{E}\,\tilde{q}_{n,k}(X_m) = \mathbf{1}\{n=0\}\,P(X_m=0) + \int q_{n,k}\,d\nu_m, \qquad k \geq 2. \tag{9}$$

(a) implies (b). Under (a), ν_m has the weakly decreasing density $x \mapsto (B_m - c)f\big(c + (B_m - c)x\big)$ on $(0,1)$. The proof that (a) implies (b) in Lemma 6 uses only that the density is weakly decreasing and integrable, so $\int q_{n,k}\,d\nu_m \geq 0$ for all $n \geq 0$ and $k \geq 1$. By (9), (b) holds, and it also holds for $n = 0$.

(b) implies (a). By (9), (b) states that $\int q_{n,k}\,d\nu_m \geq 0$ for $n \geq 1$ and $k \geq 2$. We extend this to all $n \geq 0$ and $k \geq 1$. Differentiating $p_{n,k} = p_{n+1,k} + p_{n,k+1}$ gives $q_{n,k} = q_{n+1,k} + q_{n,k+1}$. Iterating, for $N > n \geq 1$ and $M > k \geq 1$,

$$q_{n,1} = \sum_{j=n}^{N-1} q_{j,2} + q_{N,1}, \qquad q_{0,k} = \sum_{j=k}^{M-1} q_{1,j} + q_{0,M}.$$

On $[0,1]$, $q_{N,1}(x) = (N+1)x^N(1-x) - x^{N+1}$ and $q_{0,M}(x) = (1-x)^{M-1} - (M+1)x(1-x)^{M-1}$ are bounded in absolute value by 2 and 3, and they converge to zero on $(0,1)$ as $N, M \to \infty$. Since ν_m is concentrated on $(0,1)$, dominated convergence gives $\int q_{N,1}\,d\nu_m \to 0$ and $\int q_{0,M}\,d\nu_m \to 0$. Integrating the first identity therefore yields $\int q_{n,1}\,d\nu_m \geq 0$ for $n \geq 1$, and the second then yields $\int q_{0,k}\,d\nu_m \geq 0$ for $k \geq 1$.

If $\nu_m \neq 0$, Lemma 6 applied to $\nu_m/\nu_m((0,1))$ shows that ν_m has a weakly decreasing density g_m on $(0,1)$; if $\nu_m = 0$, set $g_m = 0$. Hence the restriction of λ to (c, B_m) has the weakly decreasing density $y \mapsto g_m\big((y-c)/(B_m-c)\big)/(B_m-c)$. Two such densities agree almost everywhere on the intersection of their intervals, so their right-continuous versions agree there. These versions define one weakly decreasing f on $\bigcup_m (c, B_m) = (c,\infty)$, and the restriction of λ to (c,∞) is $f(y)\,dy$ by monotone convergence. Finally, $Y \leq c$ gives $X_m = 0$, where the rows with $n \geq 1$ vanish, and $Y = +\infty$ gives $X_m = +\infty$, where all rows vanish. $\square$

For $Y \geq 0$ and $c = 0$, condition (a) states that $\lambda = w_0\,\delta_0 + f(y)\,dy + w_\infty\,\delta_\infty$ with f

weakly decreasing on $(0,\infty)$. This is the class of Example 7 without the bound B and with an additional atom at infinity. For general c, condition (a) is monotonicity above the threshold c, again without a bound.

In the application, the shaped statistics are ratios. Let $N, D \in C(K_\phi)$ with $D \geq 0$ and

$$N(\theta)^2 + D(\theta)^2 > 0 \qquad \text{for all } \theta \in K_\phi, \tag{10}$$

and let $S = N/D$, with $S = +\infty$ where $D = 0 < N$ and $S = -\infty$ where $D = 0 > N$. Then $S : K_\phi \to [-\infty, \infty]$ is continuous: near a point with $D = 0$, (10) keeps $|N|$ bounded away from zero while D tends to zero. Consequently, for all $m \geq 1$, $n \geq 0$, $k \geq 2$ and $c < B_1$, the functions $\theta \mapsto \tilde{q}_{n,k}\big((S(\theta) - c)^+/(B_m - c)\big)$ and $\theta \mapsto \tilde{q}_{n,k}\big(|S(\theta)|/B_m\big)$ are continuous on K_ϕ. By Lemma 7, a weakly decreasing density of $|S|$ on $(0,\infty)$, obtained with $Y = |S|$ and $c = 0$, and a weakly decreasing density of S above a threshold c are each represented by countably many restrictions with continuous integrands, as required by Assumption 3. Neither representation requires a bound on S or a lower bound on D. Where $D = 0$, S is infinite and every windowed row vanishes, so the shape restrictions leave prior mass on such points unrestricted.

At stage r, impose the windows $m \leq r$ and the rows with $n + k = r$ and $k \geq 2$, where $n \geq 1$ for a threshold restriction. Since $q_{n,k} = q_{n+1,k} + q_{n,k+1}$ holds pointwise and is preserved by the truncation at one when $k \geq 2$, each row of degree $n + k$ is the sum of two rows of degree $n + k + 1$ for the same window. By induction, stage r implies all rows of degree at most r for the windows $m \leq r$. The stage classes are therefore nested and their intersection is $\Pi(\phi)$, which is Assumption 4.

C Assumptions in the Oil-Market Application

This appendix checks Assumptions 1–7 for the application in Section 4, for a fixed reduced form. Assumptions 1–4 follow from the structure of the model, given that the identified set

is nonempty and, for the shape restrictions, condition (10). Assumptions 5–7 depend on the reduced form, and we give sufficient conditions for them.

Model. Let ϕ collect the reduced-form VAR coefficients and the innovation covariance matrix $\Sigma \succ 0$, let P be the lower-triangular Cholesky factor of Σ, and let Ψ_h denote the reduced-form moving-average coefficient matrices, with $\Psi_0 = I_4$. Given ϕ, the structural parameter is $\theta = Q \in O(4)$, the set of orthogonal 4×4 matrices, and the impulse responses at horizon h are $\Psi_h PQ$. The columns q_s, q_f and q_{sp} of Q correspond to the supply, flow-demand and speculative-demand shocks, and the fourth column to the residual shock. The unit vectors e_{prod}, e_{rea}, e_{price} and e_{inv} select oil production, real activity, the real price of oil and inventories. The maintained restrictions $\mathcal{A}_0$ are the weak inequalities in Table 3, and K_ϕ is the set of $Q \in O(4)$ that satisfy them. Each restriction has the form $g(Q) \geq 0$ for a function g that is linear in Q.

Table 3: Maintained sign restrictions

	Supply	Flow demand	Speculative demand	Residual
Impact responses, $h = 0$				
Oil production	$-$	$+$	$+$	
Real activity	$-$	$+$	$-$	
Real price of oil	$+$	$+$	$+$	
Inventories			$+$	
Responses at $h = 0, 1, \ldots, 11$				
Oil production (level)	$-$			
Real activity	$-$			
Real price of oil	$+$			

Notes: All signs are weak inequalities, and blank entries are unrestricted. The supply shock is an oil supply disruption, as in Kilian and Murphy (2014). Since oil production enters the VAR in growth rates, the level response of production at horizon h is $\sum_{t=0}^{h} e'_{\text{prod}} \Psi_t P q_s$. The elasticity bounds are not part of $\mathcal{A}_0$; they enter through the prior class.

In the computations, the supply shock is normalized to be expansionary, which reverses

the signs in the supply column of Table 3. The map $Q \mapsto Q\,\mathrm{diag}(-1,1,1,1)$ takes one version of K_ϕ onto the other and preserves the Haar measure. It leaves the elasticities, variance shares and historical contributions below unchanged, so the two normalizations give the same bounds.

Elasticities and events. Let $\ell = \omega_p e_{\text{prod}} - \omega_i e_{\text{inv}}$, where $\omega_p, \omega_i > 0$ are fixed sample-average weights that convert the responses of production growth and of the change in inventories into the percent change in oil use, as in Kilian and Murphy (2014). Define

$$
\begin{aligned}
N_1 &= e'_{\text{prod}} P q_f, & D_1 &= e'_{\text{price}} P q_f, & \alpha_s^F &= N_1/D_1,\\
N_2 &= e'_{\text{prod}} P q_{sp}, & D_2 &= e'_{\text{price}} P q_{sp}, & \alpha_s^S &= N_2/D_2,\\
N_3 &= \ell' P q_s, & D_3 &= e'_{\text{price}} P q_s, & \epsilon_d &= N_3/D_3.
\end{aligned}
$$

All six functions are linear in Q. On K_ϕ, the impact restrictions on the real price give $D_i \geq 0$, and those on production give $N_1, N_2 \geq 0$; N_3 can have either sign. With $c = 0.0258$, the violation events of the elasticity bounds $\alpha_s^F \leq c$, $\alpha_s^S \leq c$ and $-0.8 \leq \epsilon_d \leq 0$ are

$$
G_1 = \{N_1 - cD_1 > 0\}, \qquad G_2 = \{N_2 - cD_2 > 0\}, \qquad G_3 = \{N_3 > 0\} \cup \{-N_3 - 0.8D_3 > 0\}.
$$

Where $D_i > 0$, these are the events $\{\alpha_s^F > c\}$, $\{\alpha_s^S > c\}$ and $\{\epsilon_d \notin [-0.8, 0]\}$. Where $D_i = 0$, a nonzero numerator, that is, an infinite elasticity, counts as a violation. No ratio is evaluated, and each G_ℓ is relatively open in K_ϕ, as a preimage of $(0,\infty)$ under a continuous function or a union of two such sets.

Targets. For $h \geq 1$ and $x \in \{s, f, sp\}$, the h-month forecast-error variance share of the real price of oil due to shock x is

$$
V_{p,h}^x(Q) = \frac{\sum_{t=0}^{h-1} \left(e'_{\text{price}} \Psi_t P q_x\right)^2}{\sum_{t=0}^{h-1} e'_{\text{price}} \Psi_t \Sigma \Psi_t' e_{\text{price}}}.
$$

The denominator does not depend on Q and is at least $e'_{\text{price}}\Sigma e_{\text{price}} > 0$, so $V^x_{p,h}$ is continuous on K_ϕ. For a set $\mathcal{E}$ of months, the contribution of shock x to the change in the real price of oil over $\mathcal{E}$ is

$$H^x_{\mathcal{E}}(Q) = \sum_{t\in\mathcal{E}} \sum_{h=0}^{L} e'_{\text{price}}(\Psi_h - \Psi_{h-1}) P q_x q'_x P^{-1} u_{t-h},$$

where $\Psi_{-1} = 0$, u_t are the reduced-form residuals, set to zero before the estimation sample, and L is a fixed truncation lag. Given ϕ and the data, $H^x_{\mathcal{E}}$ is quadratic in q_x and hence continuous on K_ϕ, as are differences such as $H^f_{\mathcal{E}} - H^s_{\mathcal{E}}$. The events A_D and A_S of Section 4 are $\{v \geq 0\}$ for the continuous functions $v = V^f_{p,12} + V^{sp}_{p,12} - V^s_{p,12}$ and $v = \min\{V^f_{p,12}, V^{sp}_{p,12}\} - V^s_{p,12}$, so they are relatively closed. Posterior means of variance shares and of historical contributions are therefore continuous targets, and posterior probabilities of A_D and A_S are indicator targets, as covered by Theorem 1.

Assumption 1. With Gaussian innovations $u_t = PQ\varepsilon_t$ and $\varepsilon_t \sim N(0, I_4)$, we have $u_t \sim N(0, \Sigma)$ for every Q. Conditional on the initial observations, the likelihood therefore depends on ϕ only; see Arias and Waggoner (2018).

Assumption 2. The group $O(4)$ is compact, and K_ϕ is closed in $O(4)$ because it is defined by weak inequalities on continuous functions. With the Frobenius metric, K_ϕ is therefore a compact metric space. Configurations at which a restriction binds, such as a zero price response, belong to K_ϕ although they have Haar probability zero. Nonemptiness depends on ϕ. As discussed in Section 3, the reduced-form prior is restricted to reduced forms with a nonempty prior class, and these have $K_\phi \neq \varnothing$.

Assumption 3. The event restrictions are $\pi(G_1) \leq \kappa_s$, $\pi(G_2) \leq \kappa_s$ and $\pi(G_3) \leq \kappa_d$, with G_ℓ relatively open. The tail-only specification has no continuous restrictions. The monotonicity restrictions are the windowed rows of Appendix B, applied to α^F_s, α^S_s and $|\epsilon_d|$

with $c = 0$. Their integrands are continuous on K_ϕ if (10) holds for $(N, D) = (N_i, D_i)$, $i = 1, 2, 3$. This condition is not implied by the sign restrictions, because N_i and D_i are linear functions of the same column of Q and may vanish together on K_ϕ. It holds if and only if $\min_{Q \in K_\phi}\{N_i(Q)^2 + D_i(Q)^2\} > 0$, a minimization over a compact set. Positive values at the simulated draws do not establish it, because common zeros have Haar probability zero.

Assumption 4. The stages defined after Lemma 7, with windows $m \leq r$ and rows of degree r, satisfy it. With event restrictions only, $J = \varnothing$ and no stages are needed.

Assumption 5. Conditional on ϕ, rotations are drawn independently from the Haar measure on $O(4)$, as in Arias and Waggoner (2018), and kept if they satisfy Table 3. The retained draws are therefore iid with law ν, the Haar measure conditioned on K_ϕ, provided K_ϕ has positive Haar measure, for example because some Q satisfies all restrictions strictly. The cell condition is not automatic. A sufficient condition is $C \subseteq \mathrm{cl}(\mathrm{int}\, C)$ for every nonempty cell C, with closure and interior taken in $O(4)$: every neighborhood of a point of C then contains a nonempty open subset of C, which has positive Haar measure. For the tail-only specification with an indicator target, Corollary 1 requires only $\nu(C) > 0$ for every nonempty cell.

Assumption 6. Only the continuous restrictions must hold strictly; the event restrictions may bind, also at $\kappa_s = 0$ or $\kappa_d = 0$. The tail-only specification has no continuous restrictions. For global monotonicity, a sufficient condition is that K_ϕ contains rotations at which α_s^F, α_s^S and $-\epsilon_d$ are all positive and arbitrarily small. At stage r, choose such a Q with all three below $\min\{0.0258, B_1/(r+1)\}$, where B_1 is the smallest first window among the three statistics. The point mass at Q violates no elasticity bound, and it satisfies every row of stage r strictly because

$$q_{n,k}(x) = x^n(1-x)^{k-1}\{(n+1) - (n+k+1)x\} > 0 \qquad \text{for } 0 < x < \tfrac{n+1}{n+k+1}.$$

The conditions above are not implied by the sign restrictions. For a reported stage, Assumption 6 is certified exactly if the sampled program

$$\max_{w\in\Delta_N,\ \eta\in\mathbb{R}} \eta \quad \text{s.t.} \quad \sum_{n=1}^{N} w_n a_j(\theta^{(n)}) + \eta \le b_j \ \ (j\in J_r), \qquad \sum_{n=1}^{N} w_n \mathbf{1}_{G_\ell}(\theta^{(n)}) \le \kappa_\ell \ \ (\ell\in L),$$

has a positive value: its solution is a prior on K_ϕ that satisfies the event restrictions and every continuous restriction of stage r with margin η.

Assumption 7. It is needed only for lower posterior probabilities of A_D and A_S under monotonicity restrictions; with event restrictions only, the stages are constant. Without it, Steps 3 to 5 of the proof of Theorem 1 still show that, along a suitable schedule $r(N)$, the computed lower probability converges almost surely to $\underline{m}_\infty = \lim_r \underline{m}^{(r)}$, which satisfies $\inf_{\pi\in\Pi(\phi)} \pi(\mathrm{int}_{K_\phi} A) \le \underline{m}_\infty \le \inf_{\pi\in\Pi(\phi)} \pi(A)$. The computed value is therefore a valid, possibly conservative, lower probability. A sufficient condition for Assumption 7 uses the elasticity vector $E = (\alpha_s^F, \alpha_s^S, \epsilon_d)$, which is continuous on K_ϕ under (10). All restrictions depend on Q only through E, and the completion is unrestricted, so a prior on K_ϕ belongs to $\Pi(\phi)$ whenever it induces the same law of E as some prior in $\Pi(\phi)$. Suppose that

$$E\big(A \setminus \mathrm{int}_{K_\phi} A\big) \subseteq E\big(K_\phi \setminus A\big),$$

that is, every elasticity configuration attained on the boundary of A is also attained outside A. A measurable selection, applied on the compact sets $\{Q \in K_\phi : \mathrm{dist}(Q, A) \ge 1/j\}$ whose union is $K_\phi \setminus A$, gives a Borel map $\tau : K_\phi \to K_\phi$ with $E\circ\tau = E$, $\tau(A\setminus \mathrm{int}_{K_\phi} A) \subseteq K_\phi \setminus A$ and $\tau(Q) = Q$ otherwise. For $\pi \in \Pi(\phi)$, the prior $\pi\circ\tau^{-1}$ belongs to $\Pi(\phi)$ and assigns probability $\pi(\mathrm{int}_{K_\phi} A)$ to A. Hence $\inf_{\pi\in\Pi(\phi)} \pi(A) \le \inf_{\pi\in\Pi(\phi)} \pi(\mathrm{int}_{K_\phi} A)$, which is Assumption 7.

Reduced-form draws. All conditions are stated for a fixed ϕ. The bounds in Section 4 average conditional bounds over posterior draws of ϕ, so the conditions are needed for almost

every ϕ under the reduced-form posterior.